

**ASSESSMENT OF RAILROAD BALLAST DEGRADATION UNDER
MODIFIED DROP-WEIGHT IMPACT LOADING CONDITIONS
AND IMAGE-BASED MORPHOLOGICAL ANALYSIS**

A Thesis

Submitted to the School of Technology

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April 2026

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We hereby certify that:

- (i) The work contained in this thesis is original and has been carried out by us jointly under the guidance of our supervisor, Dr. Dinesh Gundavaram, Gati Shakti Vishwavidyalaya, Vadodara.
- (ii) The work has not been submitted in whole or in part to any other institute or university for the award of any degree, diploma, or certificate.
- (iii) We have followed the academic guidelines and ethical code of conduct provided by Gati Shakti Vishwavidyalaya in preparing this thesis.
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This is to certify that the thesis entitled “*Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions and Image-Based Morphological Analysis*”, submitted by Durgesh Kumar Legha (Roll No. 2270007) and Sandeep Patel (Roll No. 2280043) to Gati Shakti Vishwavidyalaya, is a record of bonafide research work carried out jointly by them under my supervision. The work embodied in this thesis has not been submitted, in whole or in part, for the award of any degree, diploma, or certificate at any other institute or university. In my opinion, this thesis is of the standard required for the degree of Bachelor of Technology in Civil Engineering (Specialisation: Railway Engineering), and I consider it worthy of consideration for the said degree.

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Certificate of Approval

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Durgesh & Sandeep

Abstract

This thesis addresses the critical issue of railway ballast degradation, which significantly impacts the safety, stability, and operational efficiency of railway tracks. Tailored specifically to the unique conditions of Indian Railways, the research aims to develop a deeper understanding of ballast behaviour under impact loading and to quantify the magnitude of ballast degradation using three-dimensional image processing.

At the core of the study is the modification of the European-standard drop-weight impact-test apparatus to better align with Indian Railways ballast specifications and operational conditions. The experimental set-up is calibrated so that it accurately replicates the on-field stress conditions experienced by ballast in the Indian context, providing a more relevant and reliable framework for analysis. The modified apparatus serves as a critical tool for evaluating ballast degradation under controlled yet realistic conditions.

A significant innovation in this study is the use of advanced three-dimensional image-processing techniques to evaluate morphological changes in ballast particles before and after impact testing. High-resolution structured-light scans are captured to analyse changes in particle shape, size, and texture, providing a detailed and quantitative understanding of ballast degradation. This paired before–after approach allows precise measurement of deterioration, enabling the identification of critical thresholds and patterns of degradation that are not easily detectable through traditional methods.

The outcomes of this study advance railroad-ballast degradation assessment through image-based morphology, delivering a scalable and accurate complement to conventional physical testing, with substantial implications for railway safety and maintenance.

Keywords: railway ballast; drop-weight impact loading; Indian Railways; fouling index; structured-light scanning; three-dimensional particle morphology; under-ballast mat.

Outcomes of This Project

Journal Papers

1. Legha, D., & Gundavaram, D. *Composite Railway Sleepers: A Review of Mechanical Performance, Durability and Adoption Challenges*. Railway Engineering Science (2026). (Under Review). (IF = 5.4)
2. Legha, D. K., Gundavaram, D., & Patel, S. *A Large-Scale Drop-Weight Impact-Testing Apparatus for Assessment of Indian Railway Ballast: Influence of Stress Level and Subgrade-Interlayer Configuration on Settlement and Particle Degradation*. Construction and Building Materials (2026). (Under Review). (IF = 7.2)

Patents

1. Gundavaram, D., & Legha, D. (2026). *Assessment of Railroad Ballast Degradation under Modified Drop-Weight Impact Loading Conditions*. (Indian Utility Patent) (Draft Prepared).

Conference Papers

1. Legha, D. K., Patel, S., & Gundavaram, D. (2026). *Optimization of Bitumen Content in Bitumen-Stabilized Ballast Using Flowability Characteristics*. Recent Research on Environmental Earth Sciences, MedGU 2025. Springer. (**Accepted, In Press**)
2. Legha, D., Gundavaram, D., & Mishra, G. (2026). *Global Practices of Railway Drainage Systems with Emphasis on India and Emerging Innovation*. Geotechnical Practices for Innovation and Sustainability, Proceedings of the Indian Geotechnical Conference (IGC 2025), Lecture Notes in Civil Engineering, Springer. (**Accepted, In Press**)



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List of Abbreviations

Abbreviation	Full form / Description
AAV	Aggregate Abrasion Value
AIV	Aggregate Impact Value
ANOVA	Analysis of Variance
BIS	Bureau of Indian Standards
BTP	B.Tech. Project
CAD	Computer-Aided Design
$D/d_{\max}$	Ratio of mould diameter to maximum particle size (boundary criterion)
DEM	Discrete Element Method
EN	European Norm / European Standard
FB	Flexible Base (100 mm compacted sand layer, present study)
FI	Fouling Index (after Selig & Waters, 1994)
FI _{IN}	Indian Railway Fouling Index
GSV	Gati Shakti Vishwavidyalaya
HH	High Height (450 mm drop) — legacy laboratory label
IPO	Indian Patent Office
IQR	Inter-Quartile Range
IRJ	Insulated Rail Joint
IS	Indian Standard
LAA	Los Angeles Abrasion
LH	Low Height (300 mm drop) — legacy laboratory label
Mode A	Diffuse corner abrasion (present-study classification)
Mode B	Catastrophic single-particle bulk fracture (present-study classification)
PUBM	Polyurethane Under-Ballast Mat (Getzner, present study)

Abbreviation	Full form / Description
$P_{0.075}$	Percentage finer than the 0.075 mm sieve
P_{20}	Percentage finer than the 20 mm sieve
PSD	Particle Size Distribution
RB	Rigid Base (concrete mould floor, present study)
RDSO	Research Designs and Standards Organisation
RUBM	Rubber Under-Ballast Mat (12 mm, present study)
SciPy	Scientific Python library
SEM	Scanning Electron Microscopy
SL	Structured-Light (scanning)
STL	Stereolithography (triangular surface-mesh file format)
UBM	Under-Ballast Mat
UIC	Union Internationale des Chemins de fer
UR	Unreinforced (no pad, present study)
USP	Under-Sleeper Pad

List of Symbols

Symbol	Meaning (units)
A	Surface area of a particle (mm^2)
A_{ch}	Convex-hull surface area (mm^2)
a, b, c	Semi-axes of best-fit ellipsoid (mm), $a \geq b \geq c$
C	Convexity = V/V_{ch} (dimensionless)
D	Mould internal diameter (mm)
D_{eq}	Equivalent spherical diameter = $(6V/\pi)^{1/3}$ (mm)
d_{max}	Maximum particle size in a gradation (mm)
E	Elongation = I/L (dimensionless)
F	Flatness = S/I (dimensionless)
g	Gravitational acceleration, 9.81 m s^{-2}
h	Drop height of hammer (mm or m)
I	Intermediate bounding-box dimension (mm)
K_{e}	Kinetic energy of hammer at impact = mgh (J)
L	Long bounding-box dimension (mm)
L/S	Aspect ratio (dimensionless)
$(L + I)/(2S)$	Flatness index (dimensionless)
m	Hammer mass (kg)
N	Number of blows (dimensionless)
n	Sample size (dimensionless)
p	Statistical p -value (dimensionless)
R	Roundness descriptor (dimensionless, scanner-specific)
S	Short bounding-box dimension (mm)
t	Student's t -statistic (dimensionless)
V	Volume of a particle (mm^3)
V_{ch}	Convex-hull volume (mm^3)

Symbol	Meaning (units)
σ	Nominal contact stress at specimen surface (kPa)
ΔX	Change in descriptor $X = X_{\text{after}} - X_{\text{before}}$
$\% \Delta X$	Percentage change in $X = \Delta X / X_{\text{before}} \times 100$
Ψ	Sphericity = $\pi^{1/3} (6V)^{2/3} / A$ (Wadell, 1932)

CHAPTER 1

Introduction

1.1 Background: the central role of ballast in railway track

Railway ballast is a fundamental component of conventional track structures, consisting of coarse, angular aggregates—typically crushed stone—placed beneath and around sleepers to ensure track stability and performance. Its primary functions include distributing train loads from the sleepers to the underlying subgrade, providing lateral and longitudinal resistance to track movement, and facilitating effective drainage to prevent water accumulation and loss of strength in the track foundation (Navikas et al., 2018). The mechanical behaviour of ballast is governed by factors such as particle size, shape, gradation, and strength, which influence its load-bearing capacity and resistance to degradation under repeated train loading (Koozmishi & Palassi, 2017). Over time, ballast experiences fouling and particle breakage, which can reduce its drainage capability and stiffness, leading to track settlement and increased maintenance requirements (Bassey et al., 2020). Recent studies emphasise the importance of understanding the geomechanical properties and dynamic response of ballast to improve track longevity and safety, especially under high-speed and heavy-haul conditions (Alabbasi and Hussein, 2021). Consequently, ongoing research explores alternative materials, stabilisation techniques, and advanced monitoring methods to enhance ballast performance and sustainability in modern railway systems.

The mechanical behaviour of the ballast layer is governed, ultimately, by the morphological state of its constituent grains. Freshly quarried angular particles possess interlocking micro-asperities that generate very high inter-particle friction; a well-compacted layer of such particles can carry train loads with lateral deformations of less than a millimetre per million-gross-tonne of traffic. The same layer, after a decade of service, contains particles that have been rounded by abrasion and reduced in size by fracture. The rounded assembly has lower friction, the fractured fines infill the voids and reduce drainage, and the layer settles more quickly under each successive load cycle. Track geometry degrades, the track engineer dispatches a tamping machine, and a maintenance cycle begins. Esveld (2001), Lichtberger (2005), and Indraratna et al. (2011) have all noted that the maintenance cost of ballasted track is dominated by this cycle, and that delaying the onset of measurable degradation by even a few years has first-order economic consequences for a national rail network.

Under ordinary cyclic traffic the process of ballast degradation is gradual and reasonably predictable. Under *impact* loading—the class of loading generated at rail joints, wheel flats, transition zones where the track stiffness changes abruptly, turnouts, and level crossings—the process is accelerated by impulse stresses that rise briefly to values well in excess of the contact and tensile strength of individual grains (Remennikov and Kaewunruen, 2008). Li and Davis (2005) reported accelerated settlement rates in the metre-scale vicinity of bridge abutments that were several times the rate in the adjoining line, and attributed the acceleration directly to the sharp change in support stiffness that produced momentary contact-stress amplification. It is precisely at these hot-spots of impact loading that ballast degradation is most costly, both in maintenance and in the small-probability tail of safety-relevant events.

For Indian Railways—the fourth-largest rail network in the world by route-kilometre and one of the most heavily loaded by axle-load-times-traffic—the engineering problem of ballast degradation is particularly sharp. The network operates a mix of 22.5 t axle-load freight, heavy-haul corridors with axle loads proposed to move beyond 25 t, high-density suburban passenger services, and, increasingly, semi-high-speed passenger traffic. The ballast specification governing this mix—Research Designs and Standards Organisation (2023)—prescribes angular, machine-crushed stone with a 20–65 mm size envelope. The in-service vertical contact stress at the ballast–sleeper interface is estimated to range from approximately 240 kPa under ordinary passenger loading to values approaching 500 kPa under the combined static and dynamic action of heavy freight and impact events. Any laboratory test that purports to characterise the degradation behaviour of Indian ballast must reproduce particles in this size range and stresses in this band; any test that does not, reports a number whose physical meaning is uncertain.

1.2 Problem statement

This thesis addresses two coupled gaps in Indian railway ballast assessment—gaps that reinforce each other and cannot be closed independently.

Railroad ballast, typically composed of coarse, angular aggregates, plays a critical role in providing structural support, stability, and drainage for railway tracks (Selig and Waters, 1994). Under repeated train loading, ballast undergoes degradation through mechanisms such as particle breakage, abrasion, and fouling, leading to reduced track performance, increased settlement, and higher maintenance costs (Indraratna et al., 2019). Traditional methods for assessing ballast degradation, such as sieve analysis and manual inspection, are time-consuming, labour-intensive, and often subjective, limiting their scalability and precision. The degradation process is influenced by factors including impact-loading conditions, particle morphology (size, shape, and surface texture), and environmental



Figure 1.1. Indian Railways ballasted track in service, showing the sleeper–ballast system.

variables, yet these interactions are not fully understood or efficiently quantified (Bian et al., 2020).

The primary challenge is the lack of an efficient, objective, and automated method to assess railroad ballast degradation under realistic loading conditions. Current drop-weight impact tests simulate train-induced stresses, but standard protocols do not adequately account for variations in loading intensity, frequency, or ballast morphological change over time (Koohmishi & Palassi, 2021). Moreover, traditional post-test analysis fails to capture the dynamic evolution of particle characteristics, such as angularity loss or fragmentation patterns, which are critical to understanding degradation mechanisms (Liu et al., 2022). This gap hinders the ability to predict ballast performance, optimise maintenance schedules, and design more durable track systems. There is a need for an innovative approach that integrates advanced testing conditions with analytical tools to provide a comprehensive, data-driven assessment of ballast degradation.

1.3 Objectives of the thesis

The thesis pursues five objectives:

1. To modify the standard drop-weight impact-test apparatus (based on European standards) to replicate Indian Railways ballast characteristics and loading conditions.
2. To evaluate the degradation behaviour of railway ballast under repeated impact loading using the modified testing set-up.

3. To investigate the effectiveness of different reinforcement conditions in reducing ballast degradation under impact loads.
4. To quantify morphological changes in degraded ballast particles using three-dimensional image-processing techniques (e.g. shape, angularity, and size variations).
5. To establish relationships between impact-induced degradation and morphological characteristics for improved assessment of ballast performance.

1.4 Scope

The scope of this thesis focuses on the experimental and analytical investigation of railway ballast degradation under controlled impact-loading conditions, with particular emphasis on adapting existing testing methodologies to Indian Railways requirements. The study involves modifying and calibrating a drop-weight impact-test apparatus, originally based on European standards, to simulate realistic loading conditions representative of Indian Railways, including higher axle loads and varying traffic characteristics. Laboratory experiments are conducted to evaluate ballast degradation behaviour under repeated impact loading, examining parameters such as particle breakage, gradation change, and degradation indices. In addition, the research assesses the effectiveness of selected reinforcement or stabilisation techniques in reducing ballast deterioration. A key component of the study is the application of advanced three-dimensional image-processing methods to quantify changes in ballast particle morphology—including shape, angularity, and surface characteristics—before and after degradation. The thesis further aims to establish correlations between mechanical degradation and morphological evolution of ballast particles, contributing to an improved understanding of ballast performance. The study is limited to laboratory-scale investigations, to the specific ballast type used by Indian Railways, and to impact-loading conditions, without extensive consideration of long-term field performance or environmental effects.

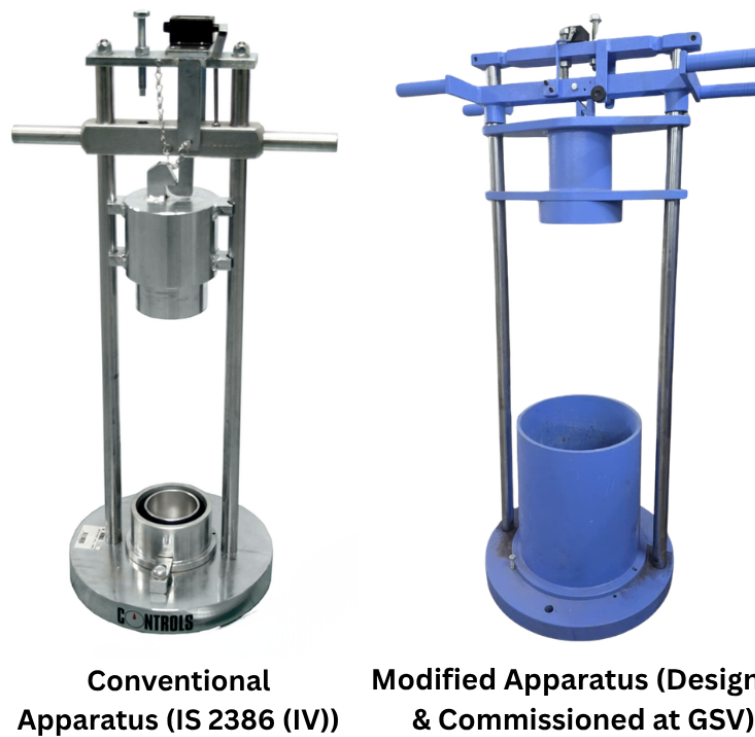


Figure 1.2. The modified impact-testing apparatus designed and commissioned at Gati Shakti Vishwavidyalaya (GSV).

CHAPTER 2

Literature Review

The literature on ballast degradation is both deep and unusually fragmented. It is deep because serious laboratory work on granular skeletons under cyclic and impact loading has been continuous since the 1960s, and it has been reinforced in the last two decades by discrete-element modelling, by field monitoring of mainline track, and—most recently—by three-dimensional imaging of individual grains. It is fragmented because these different strands have grown largely independently of one another: the cyclic-triaxial community and the impact community have different journals and different index vocabularies; the image-analysis community has only recently made sustained contact with the degradation-mechanics community; and the practitioners who set national ballast specifications have, in turn, moved at their own pace. This review is organised thematically rather than paper-by-paper in order to make sense of these strands together.

Six themes are traced in turn. [Section 2.1](#) places the ballast layer in the track system and reviews its functional requirements. [Section 2.2](#) discusses the mechanisms of ballast degradation that the thesis is ultimately concerned with. [Section 2.3](#) examines the laboratory characterisation of degradation under cyclic and impact loading, with particular attention to the apparatus and the size-effect problem that motivates [Chapter 3](#). [Section 2.4](#) reviews the use of resilient substructure inclusions—pads, mats, and compliant bases—to intercept and dissipate the impulsive energy that drives grain-scale damage. [Section 2.5](#) surveys the emergence of three-dimensional morphological characterisation, which supplies the measurement framework used in [Chapter 3](#) and applied in [Chapter 5](#). [Section 2.6](#) draws the preceding sections together and identifies the two gaps—the apparatus gap and the characterisation gap—that this thesis addresses.

2.1 The ballast layer in the railway track system

2.1.1 Functional requirements

The conventional ballasted track is a layered granular–structural system whose task is to convert the concentrated contact forces at the wheel–rail interface into diffuse, tolerable stresses at the subgrade surface. The ballast layer—a 250–350 mm deep mat of crushed, angular stone placed beneath and around the sleepers—is the principal load-distributing

element of this system. Selig and Waters (1994) enumerated its six primary functions: (i) distribution of vertical wheel loads from the sleepers to the subgrade over an area substantially larger than the sleeper footprint; (ii) restraint against lateral and longitudinal sleeper movement through inter-particle friction and interlock; (iii) absorption of some fraction of the dynamic energy imparted by passing trains; (iv) provision of rapid drainage of water away from the track formation; (v) accommodation of tamping and lining operations without structural damage; and (vi) maintenance of a resilient bearing surface that allows the rail to deflect elastically under load without plastic accumulation at the sleeper–ballast contact.

These six functions impose mutually constraining requirements on the material. High lateral restraint demands angular, machine-crushed grains with strong inter-particle interlock; high drainage demands a uniformly graded, coarse assembly with open voids; load distribution demands adequate depth and density; and long service life demands particles hard enough to resist both impact-driven fracture and cumulative abrasion. These requirements are reflected in national ballast specifications with remarkable consistency: American Railway Engineering and Maintenance-of-Way Association (2020) in the United States, UIC 719R in continental Europe, and the RDSO specification (Research Designs and Standards Organisation, 2023) in India all prescribe crushed, angular stone of comparable hardness and in a comparable 20–65 mm size band, albeit with local variations in the permissible fines fraction and in the tolerated parent-rock types.

A small but growing body of Indian-context research has begun to address the specific behaviour of RDSO-gradation ballast. Gundavaram and Hussaini (2023)—the present supervisor—demonstrated in cyclic-triaxial tests on elastomeric-polyurethane-reinforced ballast that the inclusion of a polymeric element between the ballast and the subgrade substantially reduces cumulative plastic deformation and attenuates the growth of the fines fraction. That result, on cyclic loading, motivates in part the use of the polyurethane under-ballast mat (PUBM) condition in the present impact-loading study.

2.2 Ballast degradation

2.2.1 Fouling and its engineering consequences

The engineering consequence of the grain-scale degradation mechanisms is the progressive generation of fines—particles finer than 0.075 mm in the Selig and Waters (1994) usage, or finer than the RDSO 0.075 mm sieve in the Indian adaptation. These fines infill the voids of the ballast skeleton and give rise to the condition known as *fouling*. A fouled ballast has reduced drainage capacity, reduced shear strength, a softer and more plastic response to cyclic load, and a higher rate of settlement under repeated traffic (Huang et al., 2009;

Tutumluer, 2013). The practical maintenance responses—ballast cleaning, partial renewal, and in extreme cases full deep-screening—are costly and disruptive, and the engineering motivation for slowing the rate of fines generation is direct and economic.

Field fouling has an exogenous contribution—coal and mineral-ore spillage from freight wagons, wind-blown dust, and subgrade pumping—that is not reproducible in a sealed laboratory mould. The laboratory characterisation of degradation therefore measures the endogenous contribution only. Within the laboratory, the self-generated fouling contribution is quantified by a family of bulk indices whose development and limitations are reviewed next.

2.2.2 Fouling index

The Fouling Index $FI = P_4 + P_{20}$, introduced by Selig and Waters (1994), and its metric counterpart $FI_{IN} = P_{0.075} + P_{20}$ adapted for Indian conditions by Anbazhagan et al. (2012), are weighted sums of the percentages finer than key sieves and are intended to capture the fines-generation aspect of degradation specifically, rather than the total particle-size-distribution (PSD) shift. The fouling index is dimensionless, is reported in weight percent, and is the bulk metric most directly tied to the engineering consequences that motivate ballast replacement in the field—loss of drainage, loss of shear strength, and the progressive infilling of voids that drives settlement-bowl formation at impact-prone locations. Because it isolates the fines fraction and the sub-acceptable fraction, FI_{IN} is also the most widely reported degradation metric in the Indian-ballast literature and is referenced (in its imperial form) in international ballast-fouling specifications:

$$FI_{IN} = P_{0.075} + P_{20}. \quad (2.1)$$

FI_{IN} is a scalar summary of the post-test particle-size distribution and therefore carries a structural blind spot common to any sieve-derived index. Two test specimens can produce identical FI_{IN} values while differing in the grain-scale distribution of damage: one specimen may have experienced a small amount of abrasion distributed across all its grains, another may have experienced catastrophic fracture of a few geometrically weak grains with the remainder largely untouched. If the two damage patterns happen to push comparable fractions across the 0.075 mm and 20 mm sieves, the index reports the same value. The two specimens have the same post-test PSD in those two bands, the same FI_{IN} , and completely different underlying physical states. The engineering implications—for settlement, for the spatial distribution of fines, for void fabric, for drainage—are not the same.

2.2.3 Factors affecting ballast breakage and its propagation

The rate and mode of ballast breakage are not properties of the rock alone but emerge from the interaction of intrinsic material variables with extrinsic loading and boundary variables. The intrinsic variables include the parent-rock mineralogy and grain texture, the unconfined compressive and tensile strengths, the density of pre-existing micro-cracks, and—at the assembly scale—the particle-size distribution and the particle shape (angularity, sphericity, and flatness/elongation). The extrinsic variables include the magnitude and frequency of the applied load, the stress path (monotonic versus cyclic, confined triaxial versus impact), the moisture state, the degree of confinement provided by adjacent layers, and the stiffness of the substructure on which the ballast rests. Indraratna et al. (2011) and Sun et al. (2014) summarise these dependencies and show that, holding intrinsic properties constant, the breakage index rises sharply with peak stress and with the number of load cycles, but is moderated by confining pressure (which suppresses tensile failure at point contacts) and by the introduction of a resilient layer beneath the ballast (which lengthens the contact pulse and lowers its peak).

At the grain scale the propagation of breakage is controlled by the geometry of the contact and the distribution of pre-existing micro-flaws within the parent rock. Hertzian contact theory predicts a tensile-stress lobe just below a point or near-point contact whose magnitude scales with the contact force to the one-third power; once this tensile stress exceeds the local tensile strength, a crack initiates and propagates along the path of least resistance through the grain (Lim and McDowell, 2005). Angular grains contact each other over small areas and therefore generate higher contact stresses for a given force, which is why fresher, more angular ballast fractures preferentially in the early loading history before transitioning to an abrasion-dominated regime, as already noted in [Section 2.2.1](#). Conversely, more rounded grains distribute the same contact force over a larger area, reduce the peak tensile stress, and shift the dominant degradation mode toward asperity wear. The stiffness of the substructure beneath the ballast plays an analogous role at the layer scale: a stiffer base concentrates contact forces at fewer particle–particle and particle–base contacts, while a resilient inclusion (rubber pad, under-sleeper pad, or under-ballast mat) lengthens the contact-force pulse and reduces its peak, biasing the degradation toward abrasion and away from fracture (Navaratnarajah and Indraratna, 2017; Sol-Sánchez et al., 2015).

Propagation through the assembly is governed by force-chain mechanics rather than by the average stress. Discrete-element simulations and photoelastic studies of granular media (Cundall and Strack, 1979; Tordesillas, 2007) show that load is transmitted through a sparse network of strong contacts that carry well above the mean stress, while the remaining particles bear comparatively little load. Particles that lie on a strong force chain experience contact stresses that can exceed the bulk-applied stress by an order of magnitude and are therefore the first to fracture. When such a particle splits, the chain rearranges; load is

shed onto neighbouring particles, which may themselves fracture in the next loading cycle, producing a cascade until a stable contact network re-forms. This force-chain rearrangement is the mechanism by which breakage is spatially clustered rather than uniformly distributed, and it explains the field observation that fines are often concentrated in localised pockets within an otherwise comparatively intact ballast layer.

A second statistical layer concerns the variability of breakage outcomes between nominally identical particles. Single-particle crushing tests on ballast and rockfill consistently report a wide scatter in the failure load even within a narrow size band, and the distribution is well described by Weibull statistics with a modulus typically in the range 3–6 for basaltic and granitic ballast (McDowell, 2002; Lim et al., 2004). The Weibull modulus encodes the heterogeneity of internal flaws: lower moduli indicate that a small population of weak particles will fracture early in the loading history, while the bulk of the assembly continues to degrade by abrasion. This is consistent with the bimodal damage pattern observed in laboratory specimens, where a few catastrophically split particles coexist with a majority showing only corner rounding, and it sets a lower bound on how representative any single test outcome can be of the assembly response.

2.3 Laboratory characterisation of ballast under impact loading

2.3.1 Impact-loading studies and the size-of-mould problem

Impact loading—as distinct from cyclic-triaxial loading—has received less sustained attention, despite being the loading mode that dominates at impact-prone locations in service. The seminal study of Nimbalkar et al. (2012) used a 300 mm diameter drop-weight rig with a 50 kg falling hammer and a 500 mm drop height to establish that shock mats placed beneath the ballast reduce post-test bulk degradation substantially, and that the benefit of the shock mat is greatest when the underlying support is rigid. The same authors subsequently demonstrated that recycled rubber mats achieve comparable performance at much lower cost. This line of inquiry—large-diameter mould, drop-weight hammer, ballast-scale particles—is the line that the present apparatus continues.

A series of papers by Koohmishi and co-workers has extended the drop-weight methodology in several directions. Koohmishi and Palassi (2020) investigated the morphological response of basalt, granite, limestone, and a mixed-lithology ballast under drop-weight impact and showed that the parent rock type governs the partition of damage between abrasion and splitting. Koohmishi and Azarhoosh (2021) and Koohmishi and Azarhoosh (2022) studied the effect of crumb-rubber incorporation on both fresh and recycled ballast, finding that 5–10 % rubber by volume reduces post-test bulk breakage measurably but that

the benefit saturates—and eventually reverses—beyond about 15 %. Koohmishi and Guo (2023) combined drop-weight impact data with random-forest and support-vector-regression models and demonstrated that parent rock type and nominal impact stress together explain the majority of variance in post-test bulk degradation, with rubber content contributing an additional smaller effect. Zhang et al. (2023) and the very recent Chen et al. (2026) have continued this line on ballast–rubber mixtures, with the 2026 study being the first to explicitly couple impact-loading breakage to the gradation-evolution trajectory during a multi-blow test rather than only at the endpoint.

Across all of these studies the same apparatus-related problem appears: the conventional IS/BS aggregate-impact-value test (British Standards Institution, 2010; Bureau of Indian Standards, 1963) has a 102 mm diameter mould that is far too small for RDSO-gradation ballast. Marsal (1967) established that the boundary conditions of a granular test dominate the measured response until the container diameter is at least six times the maximum particle size, and Bishop (1969) independently reached a similar conclusion from work on rock-skeleton saturation response. In a 102 mm mould with 40–65 mm particles the ratio $D/d_{\max}$ is only 1.6–2.5; one is no longer testing the ballast, one is testing the mould wall. The universal response in the published impact literature has been to develop purpose-built large-diameter rigs of 250–500 mm diameter (e.g. Ferreira and Indraratna, 2023; Koohmishi and Palassi, 2020; Nimbalkar et al., 2012)—but none of these has been designed or calibrated specifically for the RDSO gradation and the Indian stress range, and none is standardised for Indian Railways quality assurance. The present thesis fills that specific apparatus gap.

Table 2.1 makes two observations explicit. First, the dominant apparatus tradition has converged on a 250–300 mm mould, a 40–50 kg hammer, and drop heights in the 300–500 mm range—all of which are broadly consistent with the present apparatus described in Chapter 3. Second, the metrics reported are overwhelmingly sieve-based bulk metrics, with image-based morphology appearing only in the two most recent studies and in neither case as a paired before–after protocol on individually marked tracer particles. The present thesis is, to the authors’ knowledge, the first to combine a large-diameter drop-weight apparatus calibrated for RDSO gradation with a fully paired per-particle three-dimensional morphological characterisation across a factorial grid of pad type, base condition, and impact energy.

2.4 Ballast reinforcement methods

Three categories of inclusion are commonly used to reinforce or protect the ballast layer: under-sleeper pads (USPs), which sit between the sleeper and the ballast; under-ballast mats (UBMs), which sit beneath the ballast and above the sub-ballast or subgrade; and geogrids,

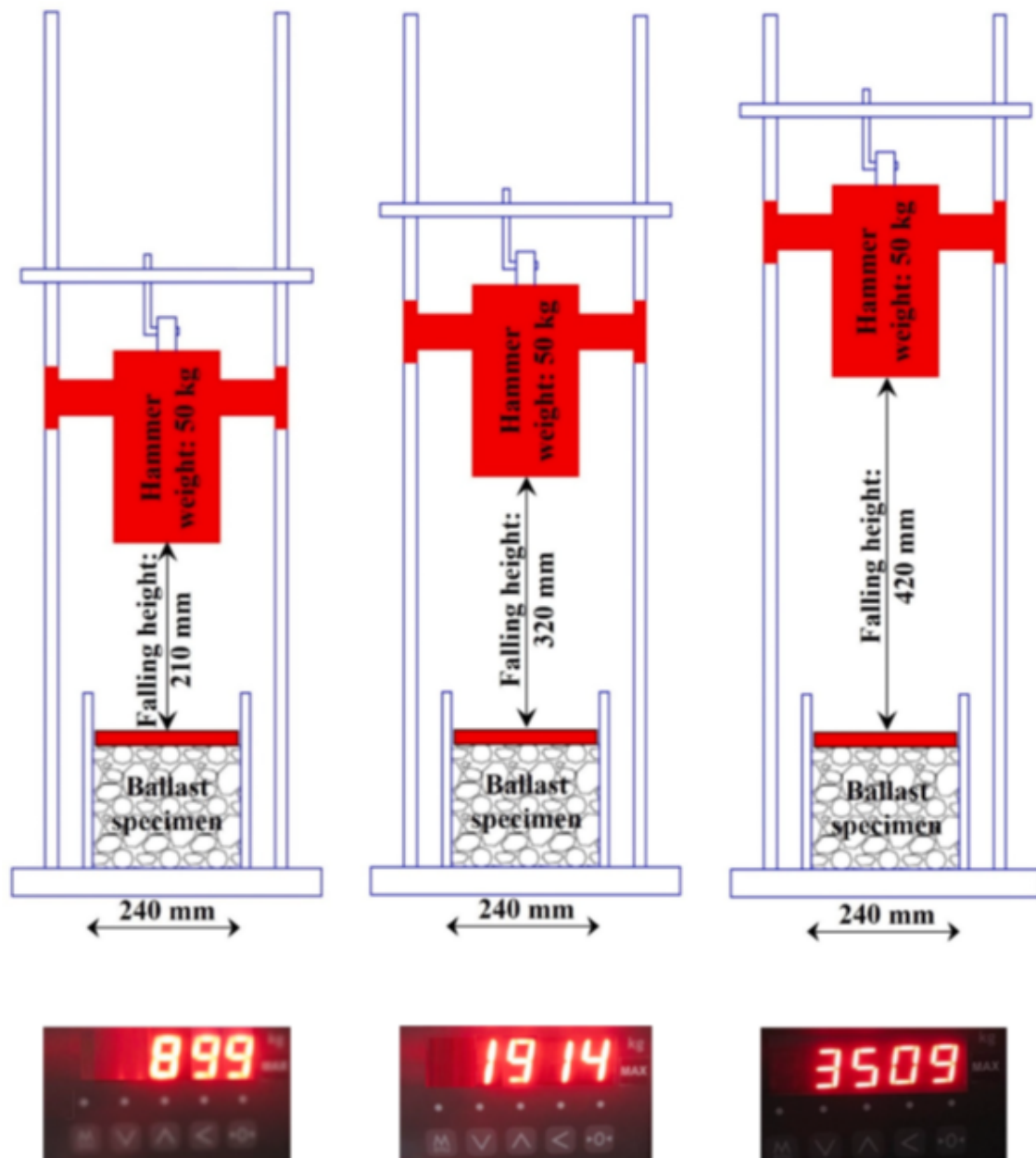


Figure 2.1. Established falling height in the impact-loading test for implementing the characterised stress levels on ballast (Koohmishi and Guo, 2023).

Table 2.1. Representative drop-weight impact studies on railway ballast: apparatus and method comparison.

Study	Mould dia.	Hammer	Drop ht.	Variables	Metrics
Nimbalkar et al. (2012)	300 mm	50 kg	500 mm	Shock mats	FI _{IN}
Koohmishi and Palassi (2020)	250 mm	40 kg	Variable	Crumb rubber, rock	FI _{IN}
Koohmishi and Azarhoosh (2021)	250 mm	40 kg	Variable	Crumb rubber, gradation	FI _{IN}
Koohmishi and Azarhoosh (2022)	250 mm	40 kg	Variable	Crumb rubber, recycling	FI _{IN}
Zhang et al. (2023)	~ 300 mm	45 kg	450 mm	Crumb rubber, energy	FI _{IN} , image
Ferreira and Indraratna (2023)	300 mm	50 kg	450 mm	Image-based morphology	2D morphology, FI
Present study	300 mm	50 kg	Variable	UR / RUBM / PUBM × FB / RB	FI _{IN} + 3D morphology

planar polymeric grids embedded within or beneath the ballast that confine particles through aperture interlock. The first two add compliance at a stiffness-mismatched interface to attenuate impact energy, whereas geogrids reinforce the ballast layer mechanically by restraining lateral spread of the granular skeleton.

Their mechanical regimes differ in kind. USPs are stiff elastomeric layers bonded to the underside of the sleeper that reduce sleeper–ballast contact stress, with static stiffness typically in the $0.10\text{--}0.30\text{ N mm}^{-3}$ band (Jayasuriya et al., 2019). UBM s are placed beneath the ballast to attenuate impact energy transmitted to the subgrade; rubber variants cover roughly $0.02\text{--}0.10\text{ N mm}^{-3}$ (Navaratnarajah and Indraratna, 2017; Ngo et al., 2019), and the polyurethane UBM (PUBM) used in the present study spans a wider product range. Geogrids, by contrast, slow lateral spread of the granular skeleton and the rate of fines generation under repeated loading. Choosing among these three is therefore a question of matching mechanism to the dominant local issue—stress reduction at the sleeper interface, energy attenuation above the subgrade, or particle confinement within the ballast itself—rather than of selecting the most compliant element available.

2.5 Three-dimensional morphological characterisation of ballast

2.5.1 From sieve-based to shape-based descriptors

The classical sieve-based characterisation of a ballast sample—its PSD and any index derived from it—does not resolve particle shape. Two assemblies with identical PSDs can differ substantially in the angularity, elongation, and surface texture of their grains, and the difference in shape is known to affect both initial strength and degradation kinetics. The classical shape-analysis literature begins with Wadell (1932), who introduced the concept of sphericity Ψ as the ratio of the surface area of a volume-equivalent sphere to the actual surface area of the particle, and with Krumbein (1941), who extended the framework to sedimentary particle populations.

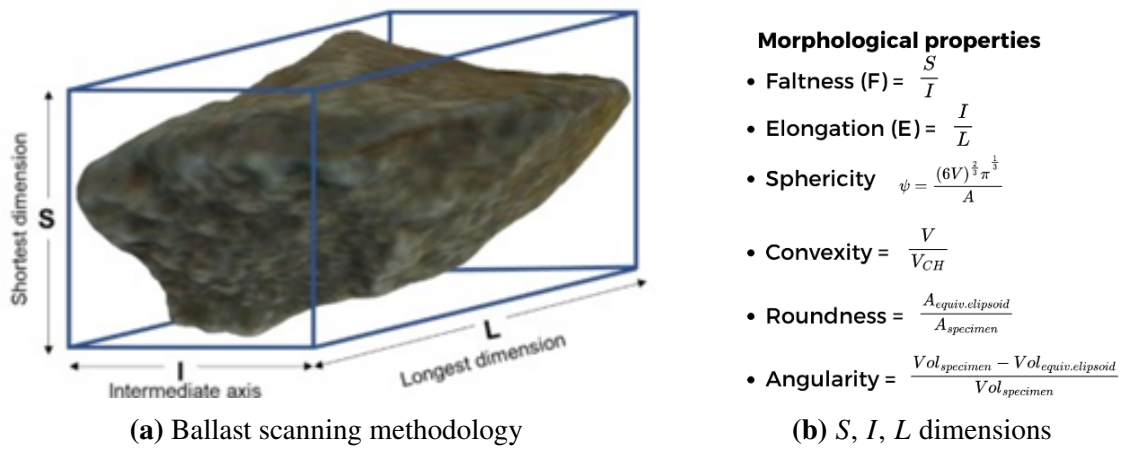


Figure 2.2. Morphological parameters and the short (S), intermediate (I), and long (L) bounding-box dimensions of a ballast particle.

2.5.2 The rise of structured-light scanning

The second generation of shape characterisation moved from manual calliper measurement to digital imaging. Sun et al. (2014) applied three-dimensional laser scanning to the size and shape of as-supplied quarry ballast and produced one of the first systematic datasets of paired particle volumes and surface areas. Qian et al. (2014) coupled image analysis to Los Angeles abrasion testing and showed that abrasion in the LAA mill follows a characteristic angularity-decay trajectory that can be tracked particle-by-particle. Guo et al. (2018) refined the angularity-index concept and, with the same LAA methodology, demonstrated that the size- and shape-dependence of degradation can be quantified at the single-particle level.

The tool-chain that supports this line of work has matured rapidly. Angelidakis et al. (2021) released an open-source shape-analysis code (SHAPE) that computes a standardised set of form and roundness descriptors from triangulated surface meshes. The open-source Python mesh-processing library `trimesh` (Dawson-Haggerty, 2019) has become the de-

facto workhorse for reading and interrogating STL files, and the scientific-computing stack NumPy and SciPy (Harris et al., 2020; Virtanen et al., 2020) supports the downstream statistical analysis. Kwunjai et al. (2023) combined a structured-light-scanning workflow with supervised machine learning to classify deteriorated ballast and demonstrated that the grain-scale shape descriptors carry the information required to distinguish fresh from service-aged ballast. Kwunjai et al. (2024) simplified the analysis pipeline while retaining the essential descriptor set.

2.5.3 Paired before–after protocols

The crucial methodological step—and the one that the present thesis is built on—is the paired before–after protocol. If the same physical grains are characterised before and after a loading event, then each grain serves as its own pre-test control and the particle-to-particle variability that is the dominant source of scatter in unpaired datasets is eliminated. The paired protocol has been used occasionally in the literature, most notably by Ferreira and Indraratna (2023) for two-dimensional image analysis of ballast under impact loading and by Bian et al. (2023) for in-situ image-aided analysis of ballast particle movement along a high-speed railway. It has not, however, been applied in combination with a large-diameter drop-weight apparatus across a factorial grid of pad type, base condition, and impact energy, which is the methodological novelty of the present study.

2.6 Synthesis and identification of gaps

Three synthesising observations emerge from the preceding five sections, and together they identify the two gaps that this thesis is designed to fill.

1. The dominant laboratory-apparatus tradition for impact loading of ballast has converged on 250–300 mm mould, 40–50 kg hammer, 300–500 mm drop-height rigs that satisfy the $D/d_{\max} \geq 6$ boundary criterion. However, no apparatus in the public record has been designed or calibrated specifically for the RDSO 20–65 mm gradation and the 275–525 kPa stress range representative of Indian Railways—the very specification and stress band that governs acceptance of ballast for the Indian network.
2. The dominant characterisation tradition—the Fouling Index, in its imperial and metric forms—is bulk in character and cannot, by construction, distinguish diffuse abrasion from catastrophic single-particle fracture. Two specimens with identical FI_{IN} values can have very different grain-scale damage patterns, with different implications for settlement-bowl formation and maintenance scheduling.

3. The emerging three-dimensional morphological tradition has demonstrated that per-particle before–after characterisation is technically feasible and methodologically sound, but has not yet been combined with a large-diameter drop-weight apparatus across a factorial grid of pad type, base condition, and impact energy—the very grid that is most relevant to the practical design of resilient inclusions at impact-prone locations on a national rail network.

CHAPTER 3

Materials and Methodology

This chapter documents the materials, the apparatus, the test protocol, and the per-particle morphological-analysis pipeline used in the experimental programme. The chapter is the longest in the thesis because the programme combines two methodological strands—bulk-index testing through a modified drop-weight impact apparatus, and paired three-dimensional morphological characterisation of individually marked tracer particles—and reproducibility of either strand requires a fully specified procedure. The chapter is organised into eight sections following the natural workflow of the programme: materials (Section 3.1), pad and base conditions (Section 3.2), tracer-particle preparation (Section 3.3), the modified drop-weight impact apparatus (Section 3.4), stress-and-energy calibration to Indian Railways field loading (Section 3.5), the blow-by-blow test protocol and bulk-index computation (Section 3.6), the image-based three-dimensional morphological-analysis pipeline (Section 3.7), and finally the statistical framework that supports the analysis of Chapters 4 and 5 (Section 3.8). The full ten-stage experimental flowchart that ties the bulk and per-particle streams together is summarised in Fig. 3.1.

3.1 Basalt ballast

3.1.1 Source and parent rock

All ballast used in the experimental programme was sourced from a single quarry supplying RDSO-specification railway ballast in the Vadodara region of Gujarat, India. A single source was used to eliminate parent-rock variability as a confounding factor between tests: any cross-test difference in degradation can then be attributed to the reinforcement configuration rather than to unrecorded mineralogical variation. The parent rock is basalt—a fine-grained, dense, aphanitic igneous rock with a dry particle density of 2.76 g cm^{-3} and a low water absorption ($< 1 \%$). These figures are consistent with the RDSO specification (Research Designs and Standards Organisation, 2023) requirement of not less than 2.60 g cm^{-3} dry density and with the parent-rock properties reported for comparable basalt in the impact literature (Koohmishi and Palassi, 2020; Zhang et al., 2023).

Approximately 25 000 kg of ballast was procured for the programme—much more than required for the nine main tests (approximately 20 kg each), for the preliminary calibration

trials run during apparatus commissioning, and for a replacement reserve. All nine tests therefore drew from the same stockpile and the same parent rock. The stockpile was kept covered and elevated from direct ground contact between tests to prevent contamination of the void-fouling baseline.

3.1.2 Particle size distribution and the RDSO envelope

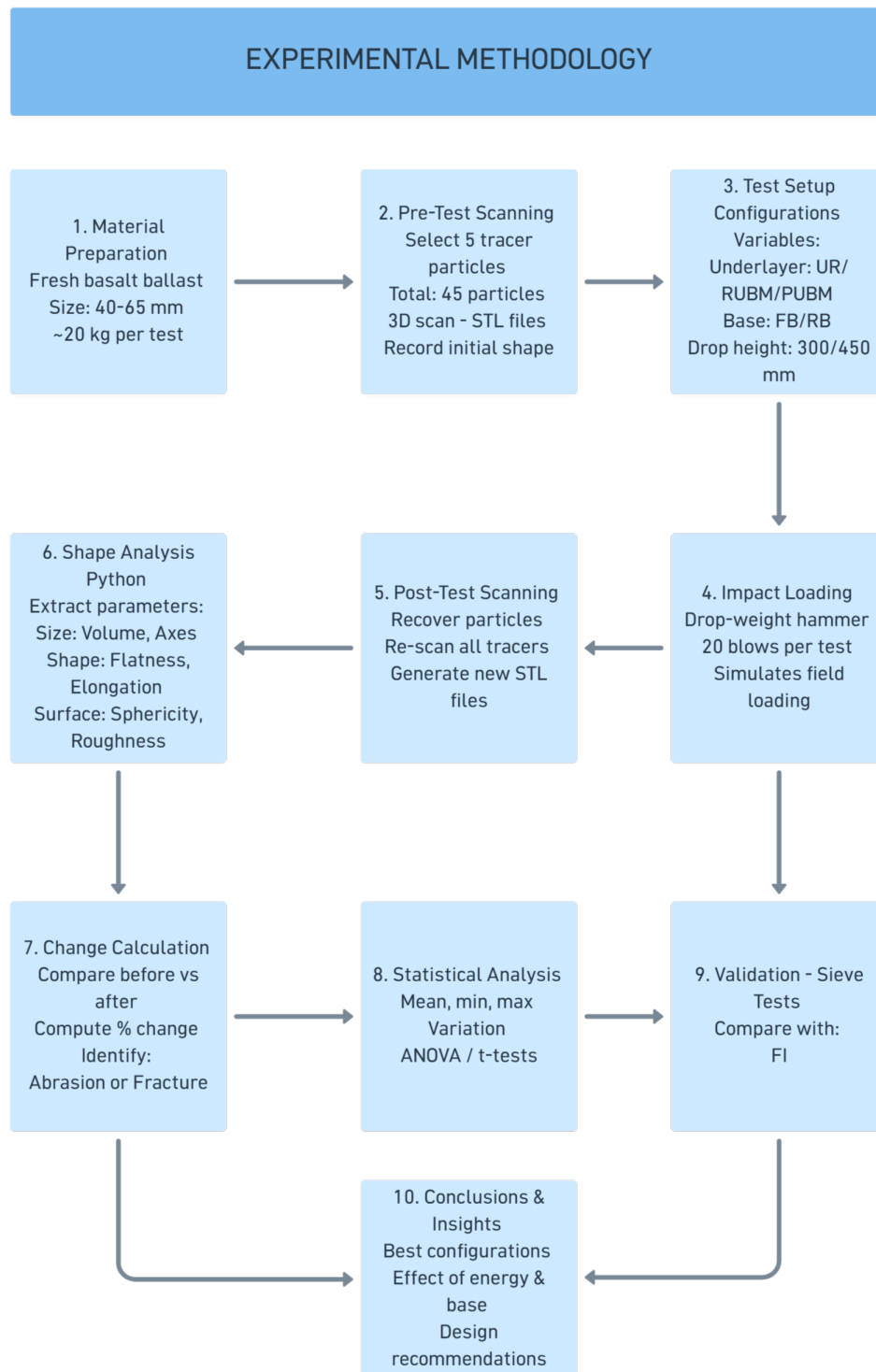
The as-supplied ballast was characterised by dry-sieve analysis using the following sieve series: 63, 50, 40, 31.5, 25, 20, 16, 12.5, 10, 4.75, 2.36, 0.6, 0.15, and 0.075 mm. Sieving was performed on sample masses of approximately 10 kg in accordance with the spirit of the RDSO procedure (Research Designs and Standards Organisation, 2023). The resulting gradation, plotted in Fig. 3.2 against the RDSO upper and lower envelope curves, shows that the ballast conforms to the specification with a small reserve against both limits—the target gradation sits close to the mid-band of the envelope throughout the 20–63 mm range. The percentage retained on the 65 mm sieve was below the 5 % maximum permitted by the specification; the percentage passing the 20 mm sieve was below 2 %, well within the 2 % maximum fines limit. The gradation is therefore centred within the acceptable window and is representative of production-quality RDSO ballast.

3.1.3 Physical and mechanical properties

The ballast was characterised for the physical and mechanical properties required by the RDSO specification. The measurements are summarised in Table 3.1. All values fall within the RDSO acceptance thresholds; the Aggregate Impact Value measured on the artificially crushed 10–12.5 mm sub-sample is reported here for completeness but is precisely the index whose limitations are set out in Chapters 1 and 2—the present thesis argues that this index cannot, by its apparatus geometry, characterise the degradation behaviour of full-size RDSO-gradation ballast, and the remainder of the programme accordingly reports the bulk fouling index FI_{IN} on the full-size specimen.

3.2 Reinforcement conditions

Three pad conditions and two base conditions are crossed in the factorial test matrix. The three pad conditions are an Unreinforced (UR) control, a Rubber Under-Ballast Mat (RUBM), and a Polyurethane Under-Ballast Mat (PUBM). The two base conditions are a Flexible Base (FB) of compacted sand and a Rigid Base (RB) of the concrete mould floor. The systematic three-letter abbreviations are introduced formally in this thesis as a replacement for the ad-hoc numbered test labels used in earlier laboratory work.



Made with  Whimsical

Figure 3.1. Experimental methodology flowchart for the ten-stage programme.

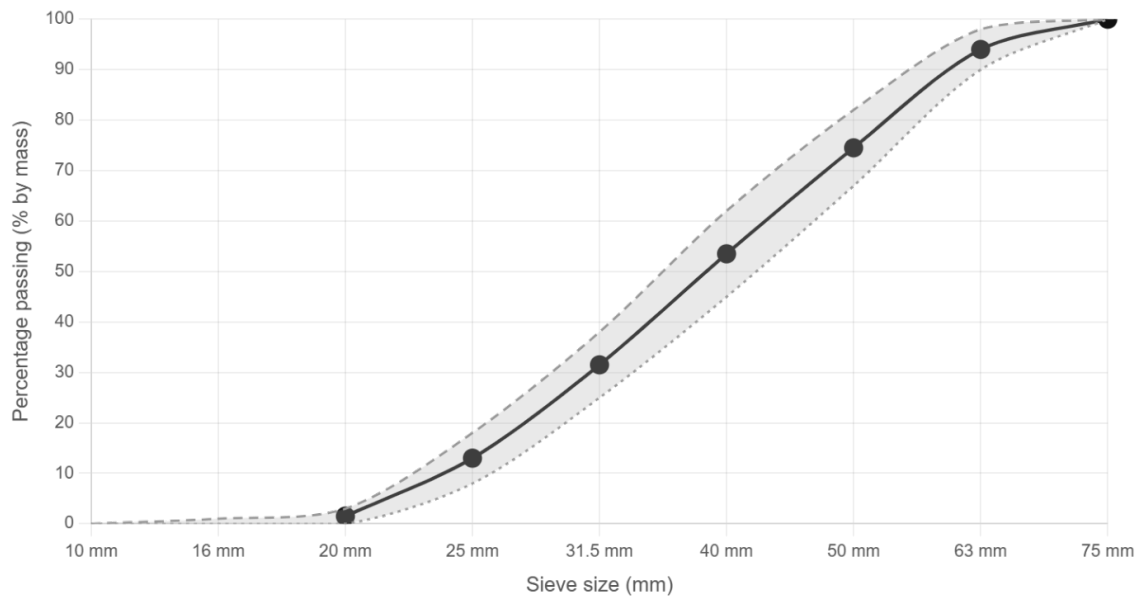


Figure 3.2. Pre-test particle-size distribution of the basalt ballast used in the experimental programme, plotted against the upper and lower envelope curves of the RDSO specification.

Table 3.1. Physical and mechanical properties of the basalt ballast used in the experimental programme.

Property	Measured	RDSO limit	Method
Dry particle density (g cm^{-3})	2.78	≥ 2.60	Gas pycnometer + weighing
Bulk (loose) density (kg m^{-3})	1460	Not specified	Bulk volume in mould
Water absorption (%)	0.72	≤ 1.0	ASTM C127 equivalent
Aggregate Abrasion Value, LAA (%)	15.4	≤ 30	IS:2386 Pt IV
Aggregate Impact Value (%)	12.1	≤ 20	IS:2386 Pt IV (conventional)



Figure 3.3. Pile of the RDSO-gradation basalt ballast used in the experimental programme (Civil Lab, GSV).

The UR configuration places no pad between the ballast fill and the base element of the mould. The UR condition serves three analytical purposes simultaneously: it is the mechanical control against which the two pad conditions are compared; it represents the in-service condition of legacy ballasted track with no substructure reinforcement; and it isolates the effect of the base condition (flexible sand versus rigid concrete) from the effect of any compliant pad placed above it. The UR condition is present in three of the nine test configurations, providing the baseline against which the pad-protected configurations are measured.

The RUBM material is a commercially manufactured 12 mm thick sheet of vulcanised natural rubber with a Shore A hardness of approximately 65 and a nominal bulk density of 1.35 g cm^{-3} . The sheet was cut to a circular disc matching the 300 mm internal diameter of the test mould and placed flat on the base element before the ballast fill. The compliance of the RUBM falls in the middle of the range reported in the literature for natural-rubber under-ballast mats (Navaratnarajah and Indraratna, 2017; Ngo et al., 2019)—sufficient to dissipate a meaningful fraction of the hammer impulse energy while stiff enough to recover elastically between blows without progressive plastic deformation of the pad itself. No RUBM specimen was reused between tests; a fresh 12 mm disc was cut for every test to eliminate any carry-over effect of cumulative pad damage.

The PUBM material is a commercial Getzner polyurethane under-ballast mat product. Getzner is a widely adopted supplier of polyurethane track-resilient elements on European, Australian, and Indian mainline and urban rail networks, and the specific variant used in this study is representative of the stiffness range in which Getzner UBM products are typically deployed beneath ballast in transition zones and impact-prone locations. The



Figure 3.4. Rubber Under-Ballast Mat (RUBM) disc: 12 mm thick natural rubber, cut to 300 mm diameter.

PUBM disc was cut to 300 mm diameter, matching the mould internal diameter, like the RUBM. Gundavaram and Hussaini (2023) has previously demonstrated, in cyclic-triaxial tests on Indian ballast, that an elastomeric-polyurethane inclusion materially reduces both cumulative plastic deformation and the rate of fines generation, and the use of PUBM in the present programme extends that line of evidence from cyclic to impact loading.



Figure 3.5. Polyurethane Under-Ballast Mat (PUBM, commercial Getzner product), cut to 300 mm diameter.

3.2.1 Base conditions (FB / RB)

The base condition—the element directly beneath the pad, or beneath the ballast if no pad is used—is the second factorial variable in the experimental design. The FB condition comprises a 100 mm deep cushion of clean, medium-grained quartz sand placed at the base of the test mould before the pad and ballast fill. The sand was oven-dried to constant mass and compacted in the mould by a standardised vibratory compaction protocol to a bulk density of approximately 1.60 g cm^{-3} . The RB condition places the ballast fill (with or without the interposed pad) directly on the concrete floor of the test mould. The concrete floor is several orders of magnitude stiffer than either the pad or the sand base and is effectively unyielding at the stress levels delivered by the hammer.

The two base conditions are chosen to bracket the stiffness range that a ballast layer might encounter in service: the FB approximates a well-compacted granular sub-ballast, while the RB approximates an unyielding structural element such as a bridge deck, a concrete track-slab foundation, or a poorly maintained subgrade that has densified below its own elastic limit. The factorial crossing of pad and base conditions allows the pad-effect contribution and the base-effect contribution to be separated cleanly, which is the analytical design that Nimbalkar et al. (2012) identified as essential for distinguishing pad performance from sub-grade compliance in laboratory shock-mat work.



Figure 3.6. Test mould prepared with the 100 mm flexible-base sand cushion (FB condition), photographed immediately before placement of the pad and the ballast fill.

3.3 Particle scanning

3.3.1 Rationale for the paired-particle design

The thesis rests on a paired-particle experimental design: the same physical grains are scanned before and after the impact-loading event, and the grain-scale damage is tracked within each particle rather than averaged across the population. The paired design is the methodological centre of gravity of the study, and its value is two-fold. First, it eliminates the particle-to-particle variability in initial shape—which in any typical ballast assembly is considerable, with volumes ranging over almost an order of magnitude across the 20–65 mm gradation—as a source of scatter in the damage signal. Second, it enables the direct attribution of each observed change in shape descriptors to the loading event, because any change recorded in the after-scan that is not also present in the before-scan of the same particle must have arisen from the hammer blows.

3.3.2 Selection criteria

Tracer particles were selected from the bulk stockpile by visual inspection and by calliper measurement. Three criteria were applied in sequence; a candidate particle had to satisfy all of them to be accepted as a tracer.

- (i) **Size:** the equivalent spherical diameter—estimated at the selection stage from the three calliper-measured principal dimensions—had to fall in the 30–65 mm range.
- (ii) **Shape:** particles with extreme flatness or elongation (aspect ratio $L/S > 3$ by initial calliper reading) were rejected, because such shapes produce preferentially high breakage under impact loading and would distort the Mode A / Mode B distribution.
- (iii) **Parent freshness:** candidates showing any visible weathering rind, iron-oxide staining, or surface cementation were rejected.

Each selected tracer particle was individually marked with a unique code. The marking protocol uses a paint marker applied to a clean, dry surface patch on each particle. The paint coat is thin (< 0.2 mm) and does not measurably affect either the volume or the surface area computed from the scans.

3.3.3 Pre-test structured-light scanning

Each tracer particle was scanned on an EinScan (Shining 3D Technology Co. Ltd., Hangzhou, China) desktop structured-light scanner. The EinScan SE was used across the programme, providing a geometric accuracy of approximately 0.1 mm and a scan resolution of approximately 0.15 mm in the automatic-turntable mode used here. The



Figure 3.7. Some of the marked tracer particles, showing the white acrylic-paint patch and the hand-written numeric identifier code.

detailed scan protocol and mesh quality-control procedures are set out in [Section 3.7](#), in the context of the morphological-analysis pipeline. In the present sub-section, pre-test scanning is described as part of specimen preparation: every tracer particle carried a complete, validated, watertight triangular surface mesh in STL format into the impact test, and the pre-test mesh serves as the reference state against which the post-test mesh is paired.



Figure 3.8. EinScan structured-light scanner (Shining 3D) in operation, with a marked tracer particle mounted on the automatic turntable.

3.4 The modified drop-weight impact apparatus

The apparatus comprises five principal components: a cylindrical cast-iron test mould of 300 mm internal diameter and 350 mm internal depth, seated on a rigid reaction base-plate; a vertical guide frame of four mild-steel columns connected by an upper crosshead; a cylindrical cast-iron hammer of 50 kg mass, sliding on the guide columns in free fall; a continuously adjustable hammer release-and-locking mechanism graduated over the 200–450 mm range of drop height; and a load-transfer assembly consisting of a 300 mm bearing plate, a compression-type 5 t strain-gauge load cell, and a hardened-steel striker plate. A mechanical blow counter and two manual operating handles complete the assembly. The mould diameter of 300 mm delivers a $D/d_{\max}$ ratio of approximately 6 against the RDSO 50 mm mean particle size, eliminating the boundary-dominance of the conventional IS:2386 Part IV apparatus (Bureau of Indian Standards, 1963) and bringing the present rig into the design envelope established by Nimbalkar et al. (2012) and the subsequent impact literature surveyed in [Section 2.3](#).

The adjustable release mechanism permits the hammer to be locked at any drop height between 200 mm and 450 mm against a graduated scale on one of the four guide columns, so that the same apparatus can deliver a range of impact energies without disassembly. The in-line load cell measures the dynamic peak force imparted through the striker plate on every blow and transmits the signal to a digital peak-force indicator; this allows each test to be characterised not only by its nominal drop-energy but by the actual force time-series, which is the principal advance of the present rig over the conventional apparatus. The mechanical blow counter provides a redundant and power-independent count of delivered blows. Detailed design drawings, material specifications, tolerance callouts, and the safety-interlock documentation are set out in the utility patent application filed for this apparatus.

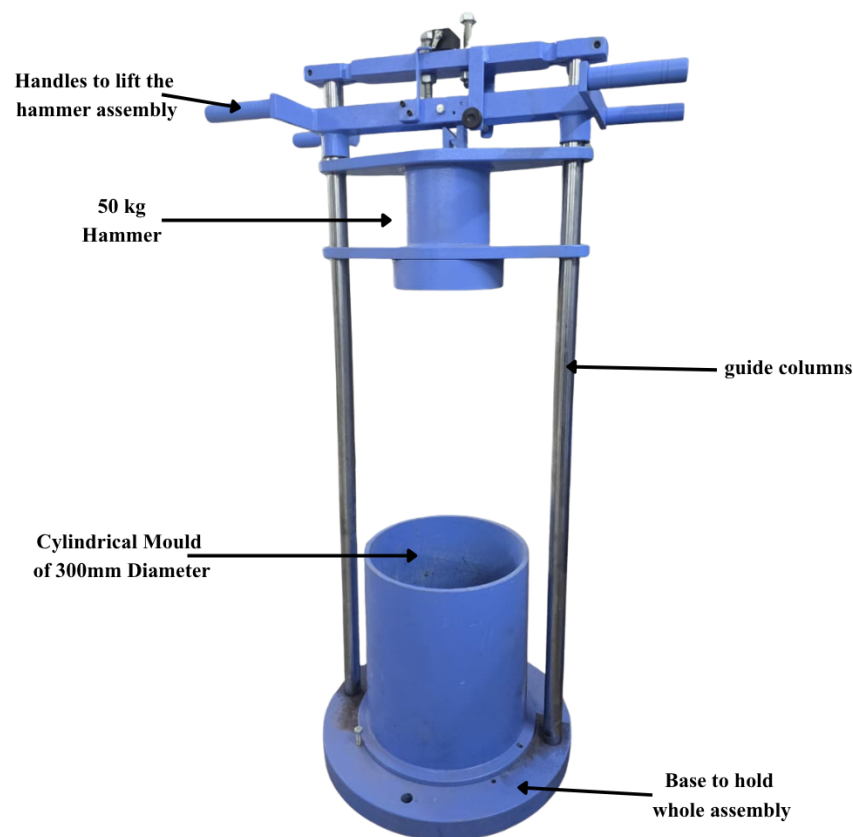


Figure 3.9. Modified drop-weight impact apparatus designed and commissioned at GSV.



Figure 3.10. Close-up of the load-transfer assembly (bearing plate, 5 t load cell, striker plate) and the digital peak-force display.

3.4.1 Comparison with the conventional Aggregate Impact Value apparatus

Table 3.2 sets out the principal specification of the modified apparatus in comparison to the conventional IS:2386 Part IV Aggregate Impact Value apparatus. Only the headline parameters are listed; the full specification is in the patent documentation.

3.5 Stress and energy calibration to Indian Railways field loading

The calibration of the apparatus to in-service stress conditions proceeds in two steps: first, the in-service stress range at the top of the ballast layer under Indian Railways axle loading is estimated; second, the hammer mass and drop-height range of the apparatus are chosen to span this in-service range when divided by the mould cross-sectional area.

3.5.1 Field stress estimation

The wheel load is distributed through the rail to several sleepers, and thence to the ballast over the effective sleeper–ballast bearing area. For a 22.5 t axle the static wheel load on each rail is $W \approx 110.25 \text{ kN}$. Only a fraction of this wheel load is transmitted to the most heavily loaded sleeper—the remainder being shared with adjacent sleepers through the bending stiffness of the rail—so the effective rail-seat load transmitted to a single sleeper–ballast contact is $Q_r \approx 27.5 \text{ kN}$. Taking an effective rail-seat bearing area below a single rail seat of $A_b \approx 1100 \text{ cm}^2 = 0.11 \text{ m}^2$, the static contact stress at the top of the ballast is

$$\sigma_{\text{static}} = \frac{Q_r}{A_b} \approx \frac{27.5 \text{ kN}}{0.11 \text{ m}^2} \approx 250 \text{ kPa.} \quad (3.1)$$

Table 3.2. Principal specification of the modified apparatus compared to the conventional IS:2386 Part IV Aggregate Impact Value apparatus.

Parameter	Conventional AIV (IS:2386 Pt IV)	Modified apparatus (present)
Mould internal diameter	102 mm	300 mm
Mould effective depth	~ 50 mm	350 mm
Hammer mass	13.5 kg	50 kg
Drop height	380 mm (fixed)	200–450 mm (adjustable)
Particle size tested	10–12.5 mm (crushed sub-sample)	20–63 mm (full RDSO gradation)
Approx. particles per test	2–4	40–80
$D/d_{\max}$ ratio	≈ 1.6 (boundary-dominated)	≈ 5
Stress simulation	Fixed / none	275–523 kPa (adjustable; maps to 22.5 t axle)
Real-time force measurement	None	5 t load cell + digital peak-force indicator
Blow counting	Manual	Integrated mechanical counter
Post-test degradation metric	AIV (% fines through 2.36 mm)	FI_{IN} + paired per-particle morphology

Applying a conservative dynamic amplification of 1.5 under ordinary mainline traffic yields $\sigma \approx 375$ kPa; at impact-prone locations, where the amplification can reach $2\times$ or higher, the contact stress rises into the 500–600 kPa range. The in-service stress window at the top of the ballast layer under Indian Railways traffic is therefore conservatively taken as approximately 250–550 kPa, with the upper end reflecting heavy-haul and transition-zone conditions.

3.5.2 Apparatus stress and energy

For the modified apparatus the nominal contact stress σ is the peak hammer force F divided by the bearing-plate area A_p , and the hammer kinetic energy at impact is $K_e = mgh$, where m is the hammer mass (50 kg), g the gravitational acceleration (9.81 m s^{-2}), and h the drop height:

$$\sigma = \frac{F}{A_p}, \quad K_e = mgh, \quad A_p = \pi \times (0.150 \text{ m})^2 = 0.0707 \text{ m}^2. \quad (3.2)$$

Using the measured peak forces reported at commissioning— $F \approx 19.4$ kN at $h = 300$ mm and $F \approx 37.0$ kN at $h = 450$ mm—the two apparatus operating points correspond to

$$h = 300 \text{ mm} : K_e = 50 \times 9.81 \times 0.300 = 147 \text{ J}, \quad \sigma = \frac{19.4 \text{ kN}}{0.0707 \text{ m}^2} \approx 275 \text{ kPa}; \quad (3.3)$$

$$h = 450 \text{ mm} : K_e = 50 \times 9.81 \times 0.450 = 221 \text{ J}, \quad \sigma = \frac{37.0 \text{ kN}}{0.0707 \text{ m}^2} \approx 523 \text{ kPa}. \quad (3.4)$$

These two operating points—275 kPa at 147 J and 523 kPa at 221 J—correspond respectively to the lower end (ordinary mainline traffic) and the upper end (heavy-haul and transition-zone conditions) of the in-service stress window. The two drop heights selected for the test programme therefore bracket the Indian Railways field-loading envelope cleanly. The adjustability of the drop-height mechanism over the full 200–450 mm range—rather than only the two operating points used in the present programme—leaves open the possibility of finer calibration to specific axle-load classes in future work.

3.6 Sample preparation

3.6.1 Specimen assembly

The nine test specimens were assembled using a single standardised protocol that produces reproducible specimens across the three pad conditions and the two base conditions. The protocol comprises eleven discrete steps, summarised here in the order of execution: (1) inspection and cleaning of the cylindrical test mould; (2) base preparation; (3) pad placement, where applicable, with fresh discs cut for every test; (4) tracer-particle placement at pre-assigned positions in the lower-middle of the ballast fill, with no tracer-to-tracer or tracer-to-wall direct contacts; (5) bulk-fill placement of the remainder of the 20 kg specimen mass in two lifts; (6) final levelling of the top surface with the load-transfer bearing-plate verification; (7) load-cell zeroing and digital-indicator verification; (8) photographic baseline record; (9) hammer positioning and locking at the specified drop height; (10) blow delivery; (11) test termination, tracer recovery, and bulk-fill collection for sieve analysis.

3.6.2 Blow-by-blow test protocol

Each of the nine tests was conducted using an identical blow-by-blow protocol. Twenty free-fall blows were delivered per test from the nominated drop height (either 300 mm or 450 mm). The choice of twenty blows is guided by two considerations. First, it is consistent with the number of blows used in comparable published drop-weight studies on ballast (Koozmishi and Palassi, 2020; Nimbalkar et al., 2012; Zhang et al., 2023). Second, the preliminary trials showed that the settlement–blow curve flattens substantially between



Figure 3.11. Assembling the test specimen in the mould (photographed without the load-cell assembly).

blow 15 and blow 20 and enters a near-stable zone beyond blow 15; twenty blows therefore captures both the transient settlement regime and the early approach to the steady state.

The twenty blows were delivered in rapid succession, with an interval of 5–8 s between consecutive blows—long enough for the operators to safely re-raise and re-lock the hammer and to record the peak-force reading, but short enough that cumulative heating of the rubber or polyurethane pad is negligible. The hammer release was triggered by two-operator manual action at the upper crosshead. After each blow the hammer was secured at the crosshead before the peak-force value was read. Axial settlement was measured at five blow milestones ($N = 0, 5, 10, 15, 20$) with a depth gauge seated on the mould rim. The digital peak-force indicator was observed and recorded for each test.

3.6.3 Test matrix and nomenclature

The nine test configurations are defined by the factorial crossing of the three pad conditions (UR / RUBM / PUBM), the two base conditions (FB / RB), and the two impact energies (300 mm and 450 mm drop heights of the 50 kg hammer, corresponding to nominal contact stresses of 275 kPa and 523 kPa as derived in [Section 3.5](#)). The nine cells listed in [Table 3.3](#) cover the factorial, with three cells (the FB-450 combinations) omitted because apparatus commissioning revealed that at the 450 mm drop height the sand cushion under the FB condition would progressively lose bearing capacity through cumulative compaction over twenty blows and would thereby confound the pad-effect signal with a base-consolidation signal.

Table 3.3. The nine test configurations of the experimental programme.

Test	Systematic label	Pad	Base	h (mm)	σ (kPa)	K_e (J)
T1	UR-FB-300	UR	FB	300	275	147
T2	RUBM-FB-300	RUBM	FB	300	275	147
T3	PUBM-FB-300	PUBM	FB	300	275	147
T4	UR-RB-450	UR	RB	450	523	221
T5	RUBM-RB-450	RUBM	RB	450	523	221
T6	PUBM-RB-450	PUBM	RB	450	523	221
T7	UR-RB-300	UR	RB	300	275	147
T8	RUBM-RB-300	RUBM	RB	300	275	147
T9	PUBM-RB-300	PUBM	RB	300	275	147

3.6.4 Post-test sieve analysis

At the end of each test the full bulk fill, together with any fines recovered from the mould floor after tracer extraction, was sieved through the same sieve series used for the pre-test gradation characterisation (63, 50, 40, 31.5, 25, 20, 16, 12.5, 10, 4.75, 2.36, 0.6, 0.15, and 0.075 mm).

3.6.5 Bulk degradation indices

A single fouling-based degradation index is computed on the post-test gradation: the Indian Railway Fouling Index FI_{IN} . Following the Anbazhagan et al. (2012) adaptation of Selig and Waters (1994),

$$FI_{IN} = P_{0.075} + P_{20}, \quad (3.5)$$

where $P_{0.075}$ and P_{20} are the weight percent passing the 0.075 mm and 20 mm sieves, respectively. The first term captures fines; the second captures the total fraction lost from the acceptable RDSO ballast gradation.

3.7 Image-based three-dimensional morphological analysis

3.7.1 Scan protocol and mesh quality control

The seven-step scan protocol applied identically to every particle in the programme—both pre-test and post-test—comprises: (1) cleaning (water wash, soft-brush scrub, distilled-water rinse, oven-dry at 105 °C); (2) mounting on a low-tack dark putty pad at the centre of the automatic turntable; (3) lighting adjustment to soft diffuse ambient; (4) first-orientation



Figure 3.12. Post-impact-test sieve analysis.

eight-view turntable scan at 45° steps; (5) reorientation of the particle to expose the previously unscanned under-side and a second eight-view scan; (6) automatic hole-filling using the scanner software; (7) export of the final merged watertight triangular surface mesh to STL format.

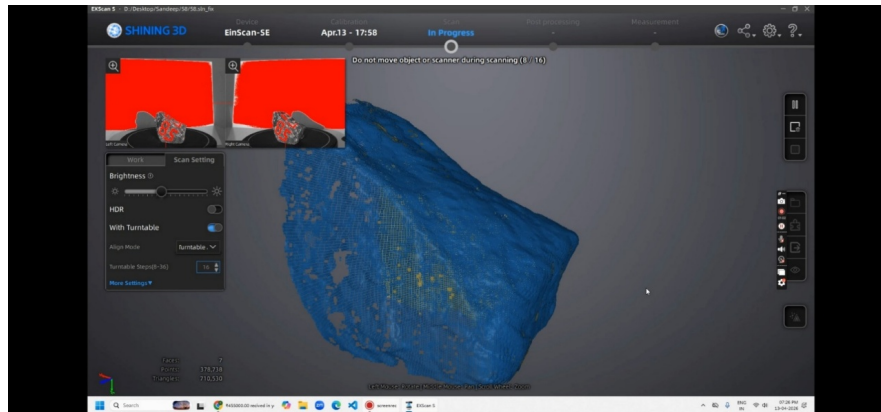


Figure 3.13. Particle scanning process used in the present study.

3.7.2 Twenty-two morphological descriptors

Twenty-two morphological descriptors are computed per mesh in three families, summarised in Table 3.4: nine size descriptors (V , A , D_{eq} , S , I , L , V_{ch} , A_{ch} , and the three ellipsoid semi-axes a , b , c); five form descriptors (flatness $F = S/I$, elongation $E = I/L$, aspect ratio L/S , flatness index $(L + I)/(2S)$, and Zingg-triangle position); and eight roundness and texture descriptors (sphericity Ψ , convexity $C = V/V_{ch}$, scanner roundness R , angularity Ang , surface-texture index $T = A/A_{ch}$, sphericity ratio $\kappa_s = D_{eq}/L$, and the cross-check pair v_{ch} and A_{ratio}). The classical Wadell sphericity (Wadell, 1932) and the Krumbein roundness concept (Krumbein, 1941) remain the two principal reporting axes for grain-scale degradation.

Table 3.4. The twenty-two morphological descriptors computed per mesh, grouped by family.

Symbol	Descriptor	Definition
<i>Size family (9)</i>		
V	Volume	mesh volume (mm^3)
A	Surface area	mesh area (mm^2)
D_{eq}	Equivalent spherical diameter	$(6V/\pi)^{1/3}$
S, I, L	Short / intermediate / long dimension	oriented bounding box
V_{ch}	Convex-hull volume	mm^3
A_{ch}	Convex-hull surface area	mm^2
a, b, c	Ellipsoid semi-axes	best-fit ellipsoid, $a \geq b \geq c$
<i>Form family (5)</i>		
F	Flatness	S/I
E	Elongation	I/L
L/S	Aspect ratio	L/S
—	Flatness index	$(L + I)/(2S)$
—	Zingg-triangle position	class from F, E
<i>Roundness & texture family (8)</i>		
Ψ	Sphericity (Wadell)	$\pi^{1/3}(6V)^{2/3}/A$
C	Convexity	V/V_{ch}
R	Roundness	scanner-specific
Ang	Angularity index	log-curvature scalar
T	Surface-texture index	A/A_{ch}
κ_s	Sphericity ratio	D_{eq}/L
ν_{ch}	Volume cross-check ratio	convex-hull ratio
A_{ratio}	Area cross-check ratio	area ratio

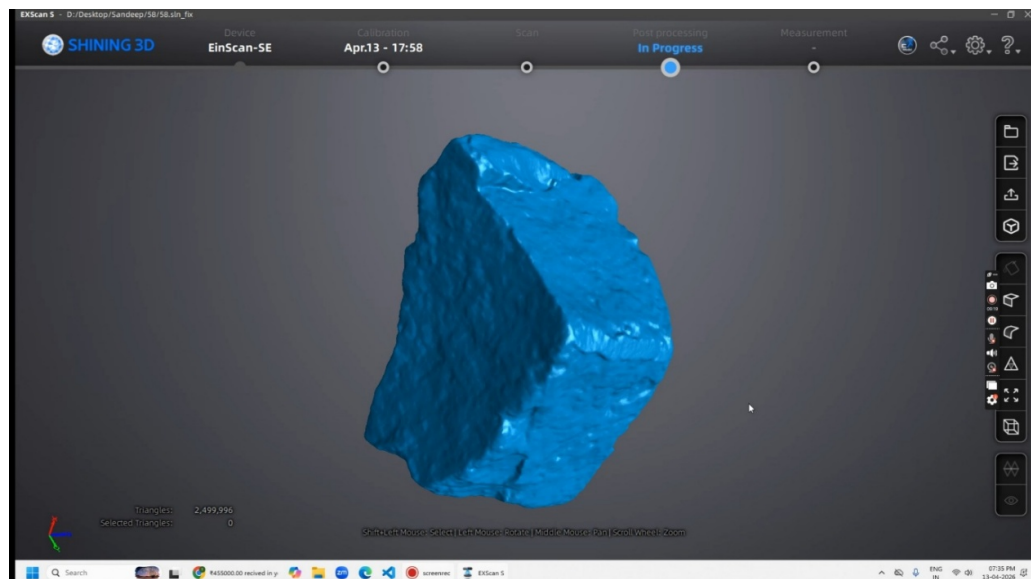


Figure 3.14. Ballast particle scanning in the EXScan software.

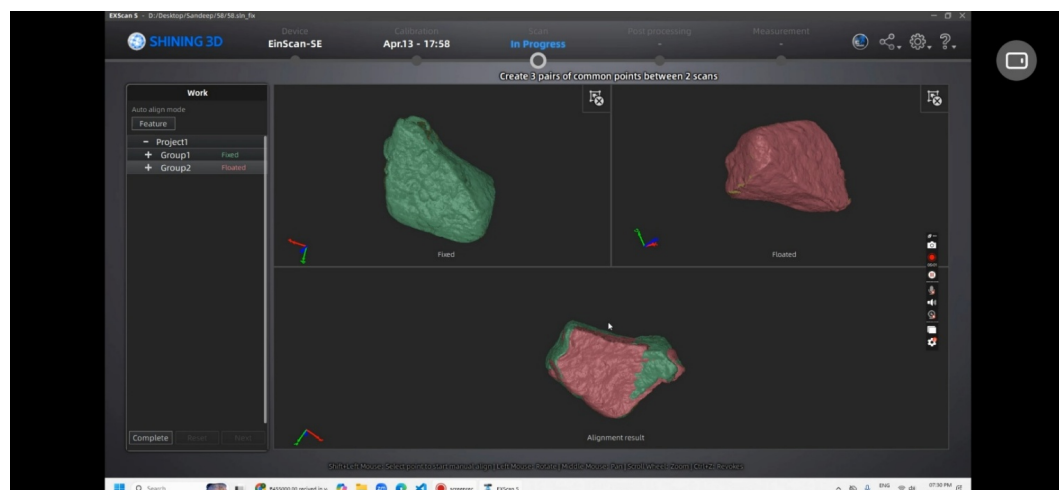


Figure 3.15. Particle aligning for multiple scans.

3.7.3 Python script

The descriptors are computed by a purpose-built Python 3 pipeline based on the `trimesh` (Dawson-Haggerty, 2019), `NumPy` (Harris et al., 2020), and `SciPy` (Virtanen et al., 2020) libraries. The pipeline executes a sequence of stages on each particle's before-and-after mesh pair: (i) load the STL; (ii) run the four-check quality-control gate; (iii) compute the size descriptors; (iv) compute the form descriptors; (v) compute the roundness and texture descriptors; and (vi) export to a multi-sheet Excel workbook with sheets for *Before*, *After*, *Changes* (After – Before), and *Percent Changes*. The complete script is reproduced in [Appendix B](#) and has been unit-tested against synthetic reference meshes (sphere, ellipsoid, cube).

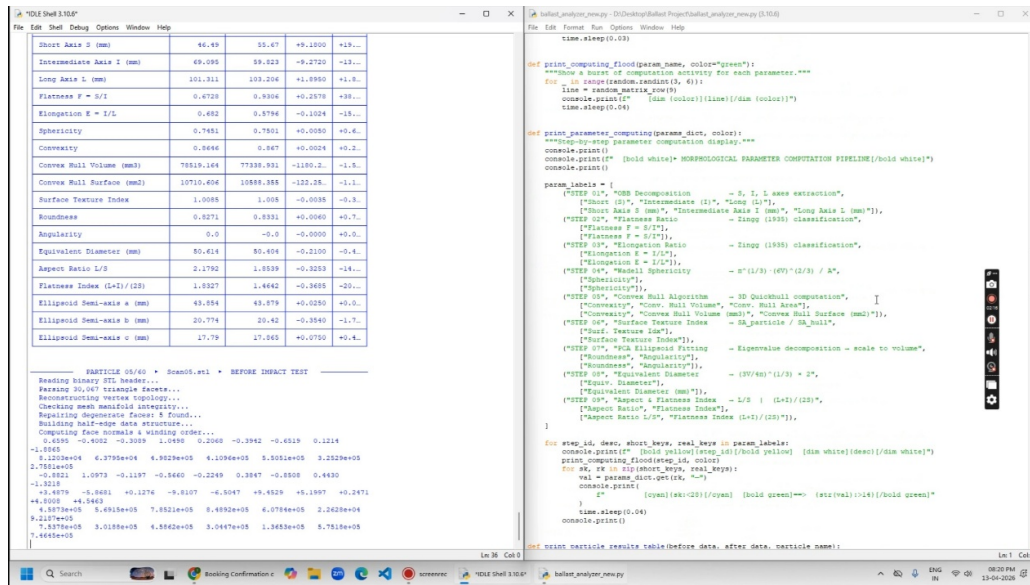


Figure 3.16. Screenshot of the running morphological-analysis script.

3.8 Statistical framework

The two characterisation streams of the programme—the bulk fouling index and the per-particle morphology—are analysed with a small, deliberately conservative set of statistical tools. This section defines them once so that [Chapters 4](#) and [5](#) can refer to them without repetition.

3.8.1 Paired comparison and significance testing

Because every tracer particle is scanned both before and after loading, each morphological descriptor X yields a paired sample $\{(X_{\text{before}}^{(i)}, X_{\text{after}}^{(i)})\}$ over the tracer population. The change for particle i is $\Delta X^{(i)} = X_{\text{after}}^{(i)} - X_{\text{before}}^{(i)}$, and the percentage change is $\% \Delta X^{(i)} =$

$100 \Delta X^{(i)} / X_{\text{before}}^{(i)}$. The null hypothesis that the loading produces no systematic change in X is tested with the two-sided paired Student's t -test on ΔX , reporting the t -statistic and the associated p -value. Pairing removes the particle-to-particle variability in initial shape and thereby sharpens the test relative to an unpaired comparison.

3.8.2 Outlier identification by the Tukey inter-quartile-range fence

A small number of particles experience catastrophic single-particle fracture (Mode B) rather than diffuse abrasion (Mode A), producing volume-change values far below the bulk of the population. These are identified objectively with the standard Tukey $1.5 \times \text{IQR}$ criterion. With Q_1 and Q_3 the first and third quartiles of the per-particle $\% \Delta V$ distribution and $\text{IQR} = Q_3 - Q_1$, a particle is flagged as an outlier (Mode B) if its $\% \Delta V$ lies outside the fence

$$\left[Q_1 - 1.5 \text{IQR}, Q_3 + 1.5 \text{IQR} \right]. \quad (3.6)$$

Particles within the fence constitute the Mode A (inlier) population; descriptor statistics are reported on the inlier population, with the outliers quoted separately rather than averaged in.

CHAPTER 4

Results of Impact-Test Analysis

This chapter reports the bulk-specimen response of the nine-configuration impact-loading programme. The chapter has three purposes. First, it documents the settlement history of the four tests in which axial settlement was monitored at five blow milestones, and uses the shake-down concept to justify the choice of twenty blows as the loading endpoint. Second, it reports the peak hammer force imparted to the specimen at the two drop heights used in the programme, confirming that the apparatus delivered the nominal energy levels reproducibly across the nine tests. Third, it presents the post-test gradation analysis of the specimens through the Indian Railway Fouling Index FI_N and examines the relationship between settlement and fouling.

4.1 Axial settlement

The cumulative axial settlement of the ballast specimen was recorded at five blow milestones ($N = 0, 5, 10, 15, 20$) in four of the nine tests: T1 (PUBM-FB-300), T3 (UR-FB-300), T5 (RUBM-RB-450), and T9 (PUBM-RB-300). The four settlement curves are plotted in Fig. 4.1, each referenced to its pre-first-blow datum so that what is shown is the net cumulative settlement.

Figure 4.2 establishes that in all four monitored tests the per-blow incremental settlement at the 15–20 interval has fallen to between approximately $0.05 \text{ mm blow}^{-1}$ and 0.5 mm blow^{-1} —between one and two orders of magnitude below the rate in the first interval. Continuing the test beyond twenty blows would therefore produce only a small additional cumulative settlement (of the order of a few millimetres at most) and would not produce qualitatively new information at the bulk level. The twenty-blow protocol of the present programme accordingly captures both the active densification regime that dominates the early blows and the entry into the shake-down regime, while keeping the test duration short enough that thermal and handling effects on the pad do not accumulate. The same twenty-blow endpoint has been used consistently in the recent drop-weight literature on ballast (Chen et al., 2026; Koohmishi and Azarhoosh, 2022; Koohmishi and Palassi, 2020; Nimbalkar et al., 2012), and the present choice is therefore consistent with established practice.

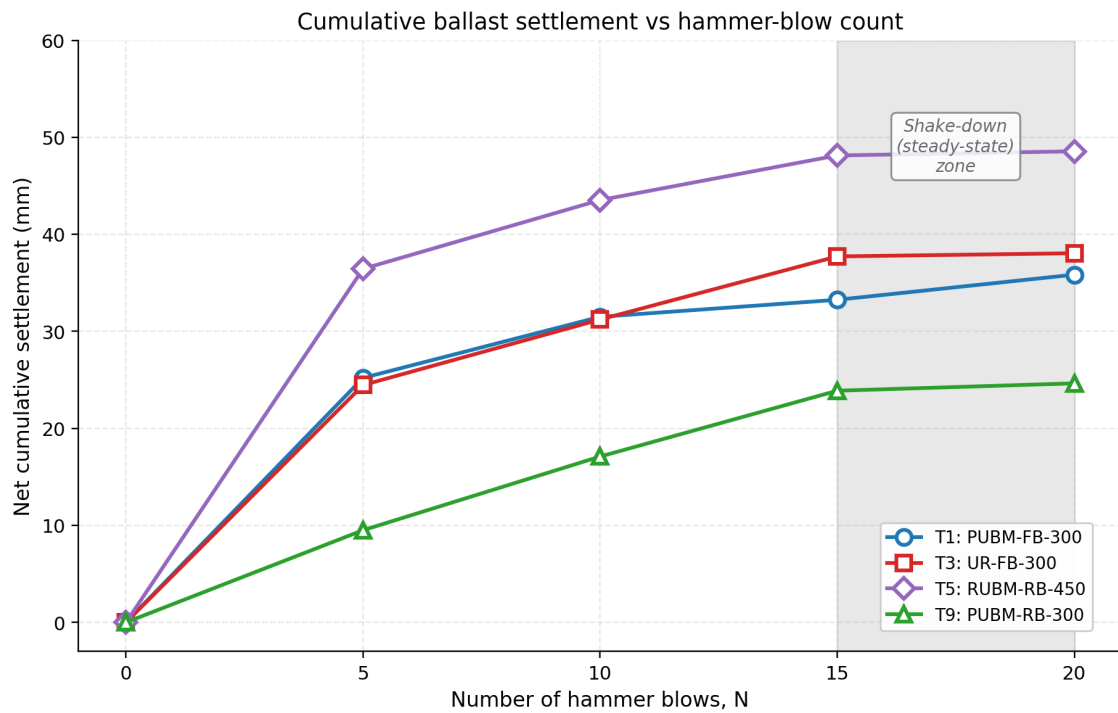


Figure 4.1. Cumulative net axial settlement of the ballast specimen as a function of hammer-blow count N for the four monitored tests. The shaded band at high N marks the shake-down (steady-state) regime, within which the incremental settlement per blow falls below approximately 0.5 mm.

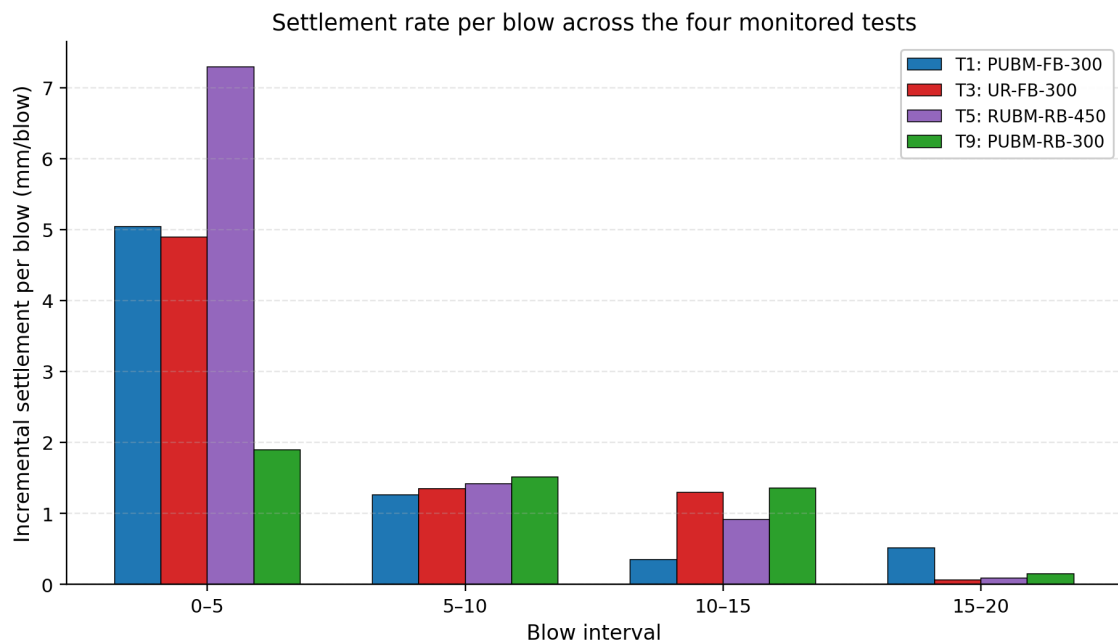


Figure 4.2. Incremental settlement per blow within each five-blow interval, for the four monitored tests. The fall in the per-blow rate from the 0–5 interval to the 15–20 interval defines the transition from active densification to the shake-down regime.

4.2 Peak hammer force per test

The in-line 5 t load cell beneath the striker plate recorded the peak compressive force imparted on every blow of every test. Across the nine-configuration programme the peak force grouped tightly by drop height: tests run at the 300 mm drop produced peak forces around 19.4 kN per blow (corresponding to a nominal contact stress of approximately 275 kPa over the 300 mm bearing-plate area), and tests run at the 450 mm drop produced peak forces around 37.0 kN per blow (approximately 523 kPa). Figure 4.3 plots the average peak force for each of the nine tests, grouped along the horizontal axis by drop height. The two-tier structure is consistent across pad type: the presence of a UR, RUBM, or PUBM pad does not appreciably change the peak force at the load cell, because the pad is below the ballast-fill column and the load cell measures the through-going force at the base of the assembly, not at the ballast surface.

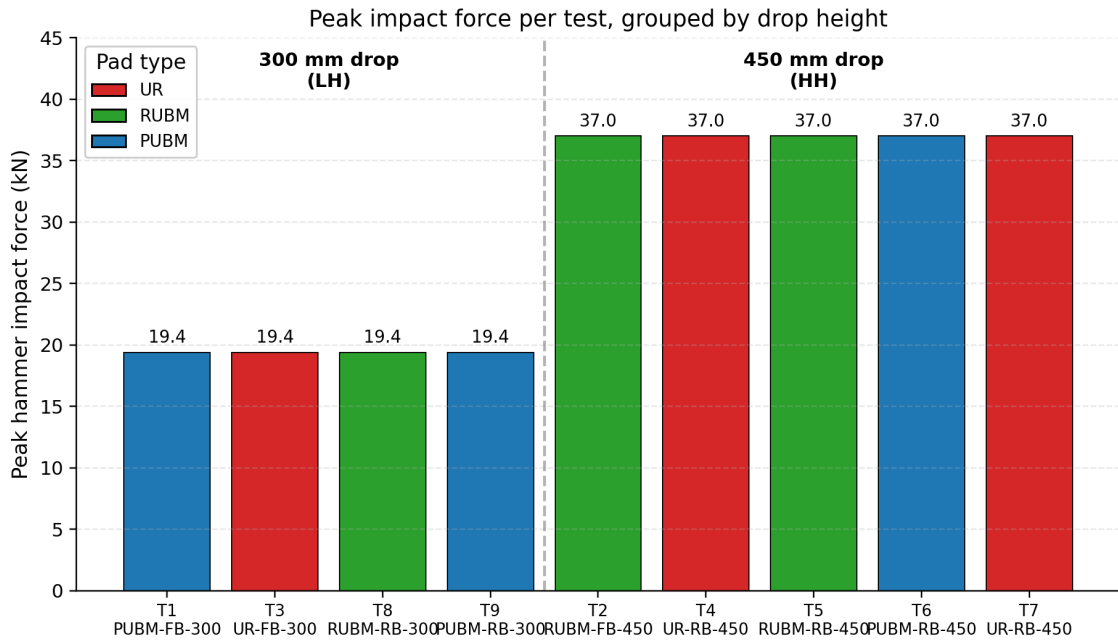


Figure 4.3. Mean peak hammer force recorded by the in-line 5 t load cell for each of the nine tests, grouped by drop height.

4.3 Indian Railway Fouling Index FI_{IN}

The post-test gradation of every specimen was determined by sieve analysis through the standard RDSO sieve series. From the resulting sieve distribution the Indian Railway Fouling Index FI_{IN} was computed as defined in Section 3.6.5 and after Anbazhagan et al. (2012):

$$FI_{IN} = P_{0.075} + P_{20}, \quad (4.1)$$

where $P_{0.075}$ is the percentage of post-test mass passing the 0.075 mm sieve (the fines fraction) and P_{20} is the percentage passing the 20 mm sieve (the fraction below the RDSO minimum acceptable particle size). The two terms have different physical interpretations: $P_{0.075}$ captures the fines that drive drainage degradation and the loss of inter-particle friction, while P_{20} captures the coarser fraction that has migrated down out of the RDSO-acceptable envelope and that, once accumulated in the field, requires either tamping or undercutting to restore. Both contribute to the overall ballast-fouling state.

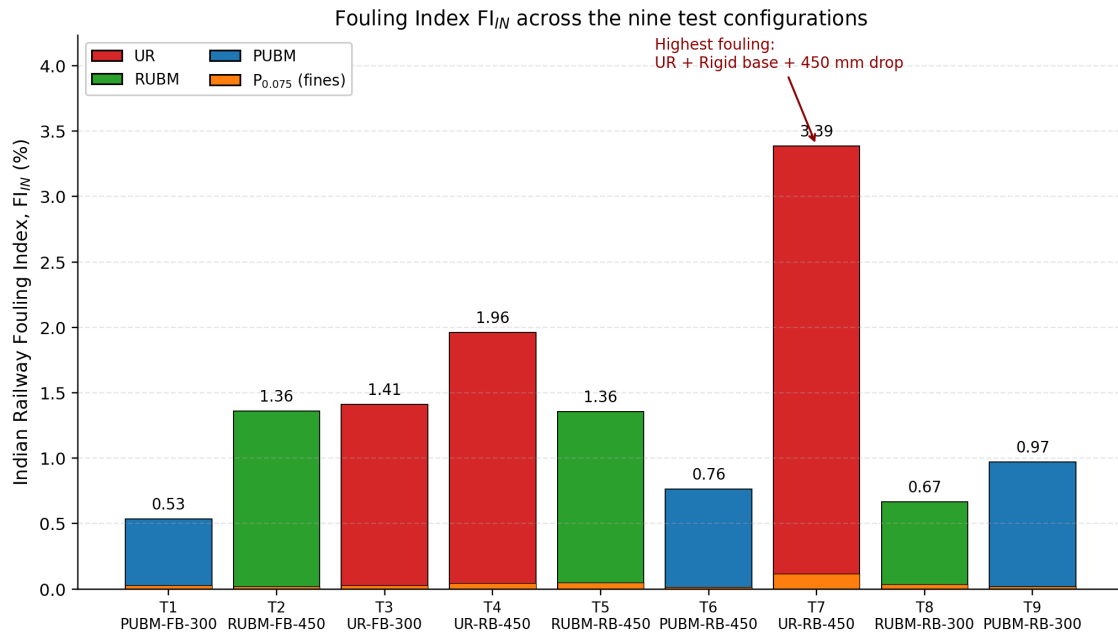


Figure 4.4. Indian Railway Fouling Index $FI_{IN} = P_{0.075} + P_{20}$ (full bar) and the $P_{0.075}$ fines contribution (overlay) for each of the nine tests.

Four observations from Fig. 4.4 and Table 4.1 are worth setting out explicitly. First, the FI_{IN} values across the nine tests span a factor of approximately 6.4—from 0.533 % at T1 (PUBM-FB-300) to 3.386 % at T7 (UR-RB-450)—which establishes that the test-configuration variables (pad type, base condition, drop height) exert a material influence on the bulk fouling response. Second, the highest-fouling configuration is T7 (UR-RB-450), the combination of no pad with a rigid base at the higher drop energy. This is mechanically the most aggressive corner of the experimental matrix—the impulsive energy is not attenuated by either a pad or a compliant base—and the result is consistent with that expectation. Third, the lowest-fouling configuration is T1 (PUBM-FB-300), the combination of a stiff polyurethane pad with a compliant sand base at the lower drop energy. The compliant sand base absorbs a substantial fraction of the impulsive energy through frictional and re-arrangement losses, and the lower drop reduces the per-blow input energy by one third relative to the 450 mm drop.

Fourth, the fines contribution $P_{0.075}$ is small in absolute terms across the programme. It contributes between 0.017 % (T6, PUBM-RB-450) and 0.118 % (T7, UR-RB-450) of

Table 4.1. Indian Railway Fouling Index FI_{IN} , the fines fraction $P_{0.075}$, and the 20 mm-passing fraction P_{20} for the nine test configurations. Values computed per the FI_{IN} definition of Section 3.6.5 from the post-test sieve data.

Test	Configuration	$P_{0.075}$ (%)	P_{20} (%)	FI_{IN} (%)
T1	PUBM-FB-300	0.027	0.507	0.533
T2	RUBM-FB-450	0.020	1.339	1.359
T3	UR-FB-300	0.028	1.384	1.412
T4	UR-RB-450	0.046	1.914	1.960
T5	RUBM-RB-450	0.049	1.307	1.356
T6	PUBM-RB-450	0.017	0.747	0.764
T7	UR-RB-450	0.118	3.268	3.386
T8	RUBM-RB-300	0.036	0.631	0.667
T9	PUBM-RB-300	0.019	0.953	0.972

the post-test mass—only a small fraction of the total FI_{IN} , which is dominated by the P_{20} term. This indicates that the dominant bulk-level damage mechanism in the present programme is the migration of particles below the 20 mm RDSO acceptable lower bound (i.e. fragmentation that produces 16 mm and smaller daughter particles), rather than the production of fines through abrasion of asperities. The grain-scale diagnostic of Chapter 5 revisits this distinction directly.

A further pattern that emerges from Table 4.1 is the relative ranking of the three pads on each base. On the rigid base at the 450 mm drop the PUBM pad (T6, $FI_{IN} = 0.764\%$) outperforms both the RUBM pad (T5, 1.356 %) and the no-pad control (T7, 3.386 %). On the rigid base at the 300 mm drop both pads outperform the comparable unreinforced configuration (T8 RUBM 0.667 %, T9 PUBM 0.972 %). On the flexible base at the 300 mm drop the PUBM pad (T1, 0.533 %) outperforms the no-pad control (T3, 1.412 %), again confirming that adding a polyurethane mat below the ballast reduces bulk fouling. The pad–base coupling pattern therefore points consistently toward the PUBM as the most effective fouling-reducer in the matrix tested.

4.4 Settlement–fouling correlation

A natural question that follows from reporting both settlement and FI_{IN} is whether the two metrics are correlated across the subset of tests for which both are available. A positive correlation would support the hypothesis that fouling-driven fines generation contributes to the cumulative axial settlement under repeated impact loading; a weak or absent correlation would suggest that the two metrics read different aspects of the response. The correlation is therefore evaluated on the four-test settlement subset.

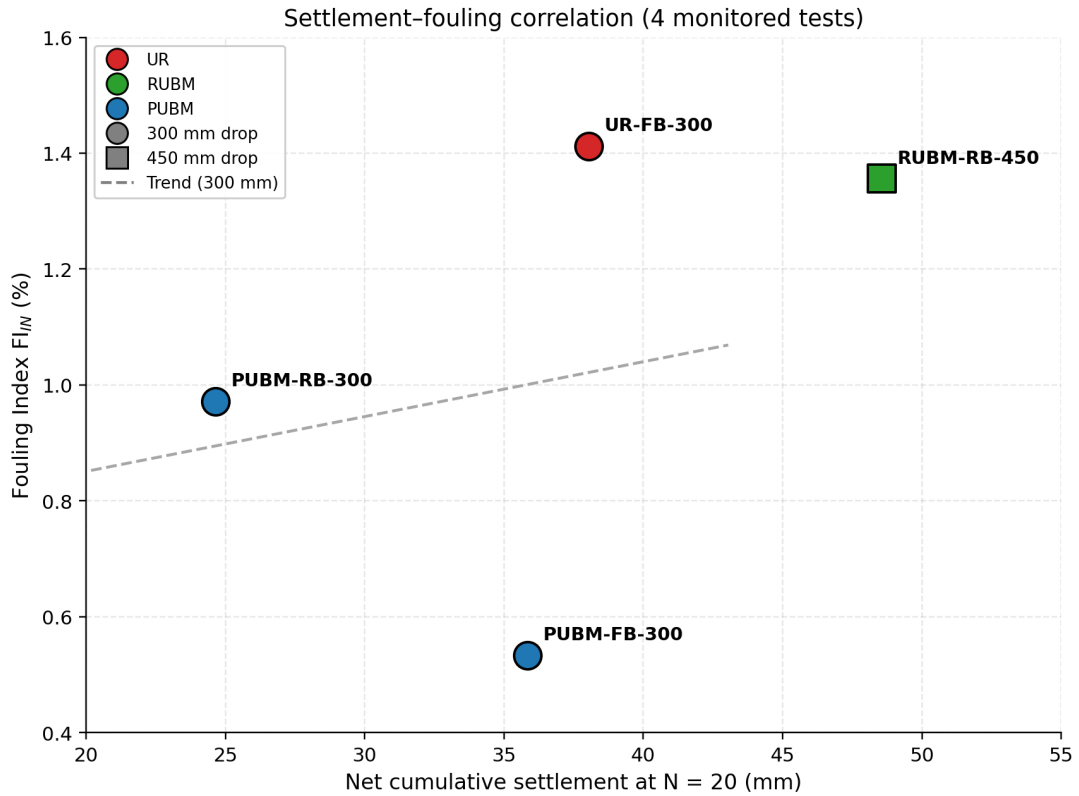


Figure 4.5. Indian Railway Fouling Index FI_{IN} plotted against the net cumulative settlement at $N = 20$ for the four monitored tests.

Two observations follow from Fig. 4.5. First, within the three low-energy 300 mm tests (T1, T3, T9) a weakly positive trend exists between settlement and fouling: the test with the largest settlement (T3, UR-FB-300 at 38.0 mm) is also the one with the largest FI_{IN} (1.412 %), and the test with the smallest settlement (T9, PUBM-RB-300 at 24.6 mm) has an intermediate FI_{IN} (0.972 %). The slope is shallow and the scatter is wide given only three points, so this trend should be read as directional rather than predictive. Second, the high-energy test T5 (RUBM-RB-450) lies clearly off the 300 mm trend: it has the largest settlement of the four (48.6 mm) but a comparatively modest FI_{IN} (1.356 %). Settlement and fouling are therefore not interchangeable bulk metrics; they index different aspects of the underlying degradation.

The decoupling is mechanistically intelligible. Settlement reflects axial deformation from particle rearrangement and void closure, which need not involve fragmentation. Fouling, by contrast, requires actual particle breakage and is governed by peak contact force and parent-rock brittleness. Thus T5 settles substantially without much fragmentation, while T7 fragments without the largest settlement.

4.5 Discussion

The bulk findings of [Sections 4.1 to 4.4](#) are broadly consistent with the published drop-weight impact literature on ballast. The shock-mat-on-rigid-base condition explored by Nimbalkar et al. (2012) produced a similar pattern of reduced bulk degradation when a compliant pad was placed beneath the ballast on a rigid support, and the natural-rubber and recycled-rubber mats in that study delivered the largest improvement on the stiffer support—directionally consistent with the present finding that PUBM substantially outperforms the unreinforced configuration on the rigid base. The crumb-rubber-modified ballast studies of Koohmishi and Azarhoosh (2021, 2022) and Zhang et al. (2023) reported fines-generation values in the same order of magnitude as the present $P_{0.075}$ results at comparable energy, and Koohmishi and Palassi (2020) reported similar fragmentation severity on basalt ballast under matched drop-weight loading. The shake-down framework used to justify the twenty-blow protocol in [Section 4.1](#) follows the treatment of Werkmeister et al. (2001) for granular materials and is the same framework used in the impact-loading context by Nimbalkar et al. (2012).

Three patterns visible at the bulk level are worth recording before moving to [Chapter 5](#). First, the PUBM-on-rigid-base configurations T6 and T9 outperform the RUBM and UR equivalents at both drop heights, suggesting that a stiffer polyurethane mat is more effective than a softer rubber mat at attenuating impulsive loading on a rigid support. Second, the highest- FI_{IN} configuration is T7 (UR-RB-450), the no-pad rigid-base high-energy combination—the mechanically most aggressive corner of the experimental matrix and the one most representative of the unprotected rigid-substructure conditions encountered at bridge approaches and welded-rail transition zones in service. Third, the settlement and fouling responses are not interchangeable: the test with the largest settlement (T5) is not the test with the largest fouling (T7). Both metrics are needed to characterise the bulk response, and even together they leave open the question of which grain-scale processes drive each metric in each configuration. That is the question addressed in [Chapter 5](#).

CHAPTER 5

Results of Morphological Analysis

This chapter reports the results of the paired three-dimensional morphological analysis of the 45 tracer particles—five per configuration across the nine configurations of the experimental programme. The chapter is organised in three sections. [Section 5.1](#) reports the per-particle volume-change distribution and identifies a small set of outlier particles using the Tukey inter-quartile-range criterion described in [Section 3.8.2](#). [Section 5.2](#) reports the paired evolution of four principal shape descriptors—sphericity, convexity, angularity, and roundness—before and after the impact loading. [Section 5.3](#) examines the global pattern of mean change across all twenty-two morphological descriptors using two heat maps.

5.1 Per-particle volume change and outlier identification

Across the 45 particles the $\% \Delta V$ distribution has a first quartile of -1.21% , a third quartile of -0.09% , and an inter-quartile range (IQR) of 1.12% . Applying the standard Tukey $1.5 \times \text{IQR}$ fence of [Equation \(3.6\)](#) to identify outliers yields a lower fence at -2.89% and an upper fence at 1.59% . Six particles fall below the lower fence and are flagged as outliers; no particle falls above the upper fence.

The Mode A / Mode B classification used to describe these two populations in earlier documentation is retained as a labelling convenience throughout this chapter—Mode A refers to the 39 inlier particles (within the Tukey fence) and Mode B refers to the six outlier particles outside it—but no further mechanistic content is read into the labels themselves. The remainder of the chapter reports the morphological response of the inlier population and treats the outliers as a separately identified set whose contribution to each descriptor is quoted but not averaged into the inlier statistics.

5.2 Shape evolution: sphericity, convexity, angularity, and roundness

Angularity and roundness are the two principal descriptors of the sharpness of corner-and-edge features on a particle, while sphericity and convexity integrate over the whole grain.

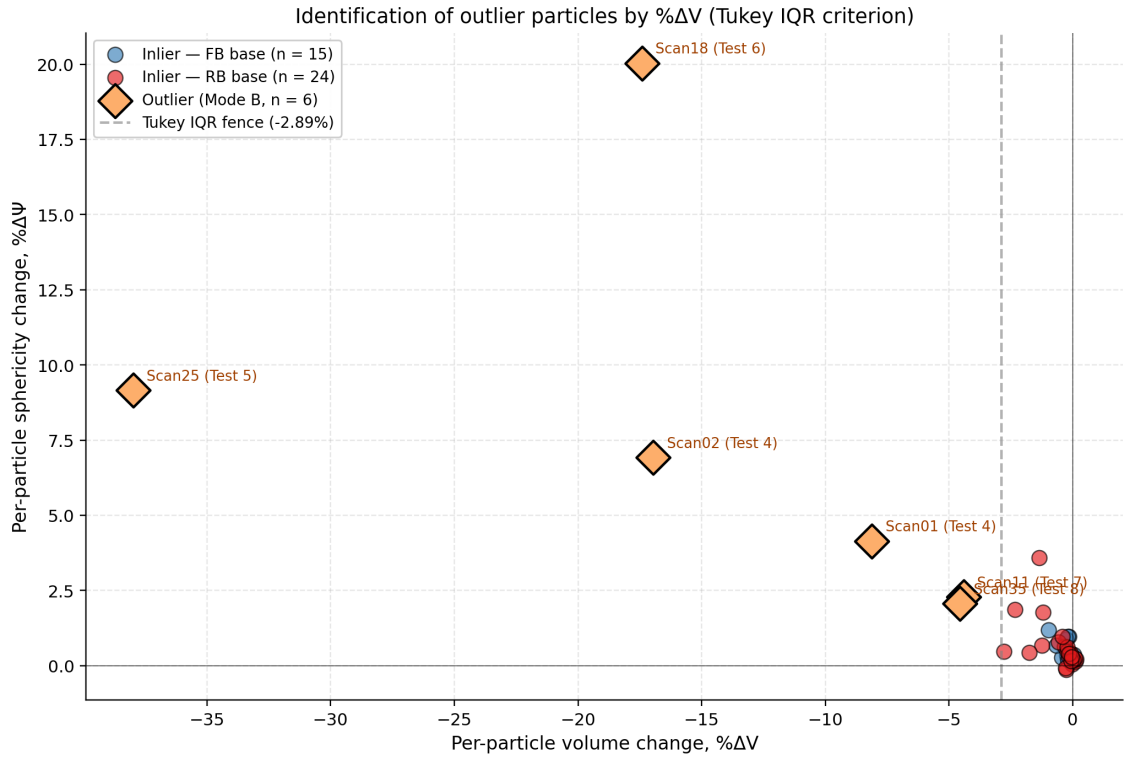


Figure 5.1. Per-particle relative volume change $\% \Delta V$ (horizontal axis) plotted against per-particle relative sphericity change $\% \Delta \Psi$ (vertical axis), for all 45 tracer particles, with outliers (Mode B) identified.

Table 5.1. Per-test summary of the volume-change response. n_{in} is the number of inlier (Mode A) particles, n_{out} the number of outlier (Mode B) particles; the mean and median $\% \Delta V$ are computed on the inlier population.

Test	Configuration	n_{in}	n_{out}	Mean $\% \Delta V$	Median $\% \Delta V$
T1	PUBM-FB-300	5	0	-0.18	-0.20
T2	RUBM-FB-450	5	0	-0.10	-0.09
T3	UR-FB-300	5	0	-0.44	-0.44
T4	UR-RB-450	3	2	-0.87	-1.24
T5	RUBM-RB-450	4	1	-0.84	-0.25
T6	PUBM-RB-450	4	1	-0.30	-0.07
T7	UR-RB-450	4	1	-0.71	-0.45
T8	RUBM-RB-300	4	1	-0.58	-0.04
T9	PUBM-RB-300	5	0	-0.18	-0.15

Both classes are expected to evolve toward smoother values under abrasive impact loading, though the corner-sensitive descriptors depend on the subtler local curvature of the surface mesh rather than the global aspect ratio.

Figure 5.2 plots the angularity index before versus after the test for all 45 particles; the angularity values reported here are negative scalars, so less-negative (closer to zero) values indicate smoother corners. The mechanistic interpretation is that the angularity descriptor is sensitive to local surface curvature and is therefore strongly affected by which corner is preferentially abraded on a given particle. A particle with two sharp corners may lose only one of them under twenty blows; the remaining corner is captured by the maximum-curvature feature in the post-test scan and the descriptor records little change, or even a small increase if the remaining corner becomes more pronounced relative to the now-smoothed neighbour. This within-particle redistribution of curvature is not visible at the level of the global sphericity—which integrates over the whole particle—but is captured directly by angularity. The scatter in Fig. 5.2 therefore does not contradict the smoothing-by-abrasion picture established by sphericity; it adds the observation that the smoothing is locally selective, not uniformly distributed across all corners of all particles.

The sphericity—the classical Wadell descriptor (Wadell, 1932)—is the single most informative shape descriptor for diagnosing corner abrasion. A particle that loses its sharp corners without otherwise changing shape becomes more spherical. Figure 5.3 plots the paired sphericity— Ψ_{after} on the vertical axis against Ψ_{before} on the horizontal axis—for all 45 tracer particles. The unit-slope line $\Psi_{\text{after}} = \Psi_{\text{before}}$ is drawn for reference; points above this line indicate a sphericity increase, points on it indicate no change, and points below it a sphericity decrease.

The systematic displacement of the points above the unit-slope line is the single most consistent finding of this chapter. The mean paired sphericity change across all 45 particles is $\Delta\Psi = +0.0100$, and the paired t -test against the null of zero mean change yields $t = 3.23$, $p = 0.002$. Restricting attention to the 39 inlier particles only, the mean change is $\Delta\Psi = +0.0041$ and the t -statistic is 6.05 with $p < 10^{-6}$. The smaller mean for the inliers reflects the fact that the outlier particles, after their fragmentation, are individually more spherical than they were before; their removal slightly reduces the average rise but, more importantly, sharpens the significance of the rise considerably because the within-population variance falls. The conclusion is that sphericity rises in essentially every particle regardless of configuration. Corner abrasion is therefore a universal response to the drop-weight impact protocol, and no configuration in the matrix—including the best-performing T1 (PUBM-FB-300)—eliminates it. This is consistent with the abrasion-driven rounding observed in cyclic loading by Qian et al. (2014) and Guo et al. (2018), and in drop-weight impact loading by Koohmishi and Palassi (2020).

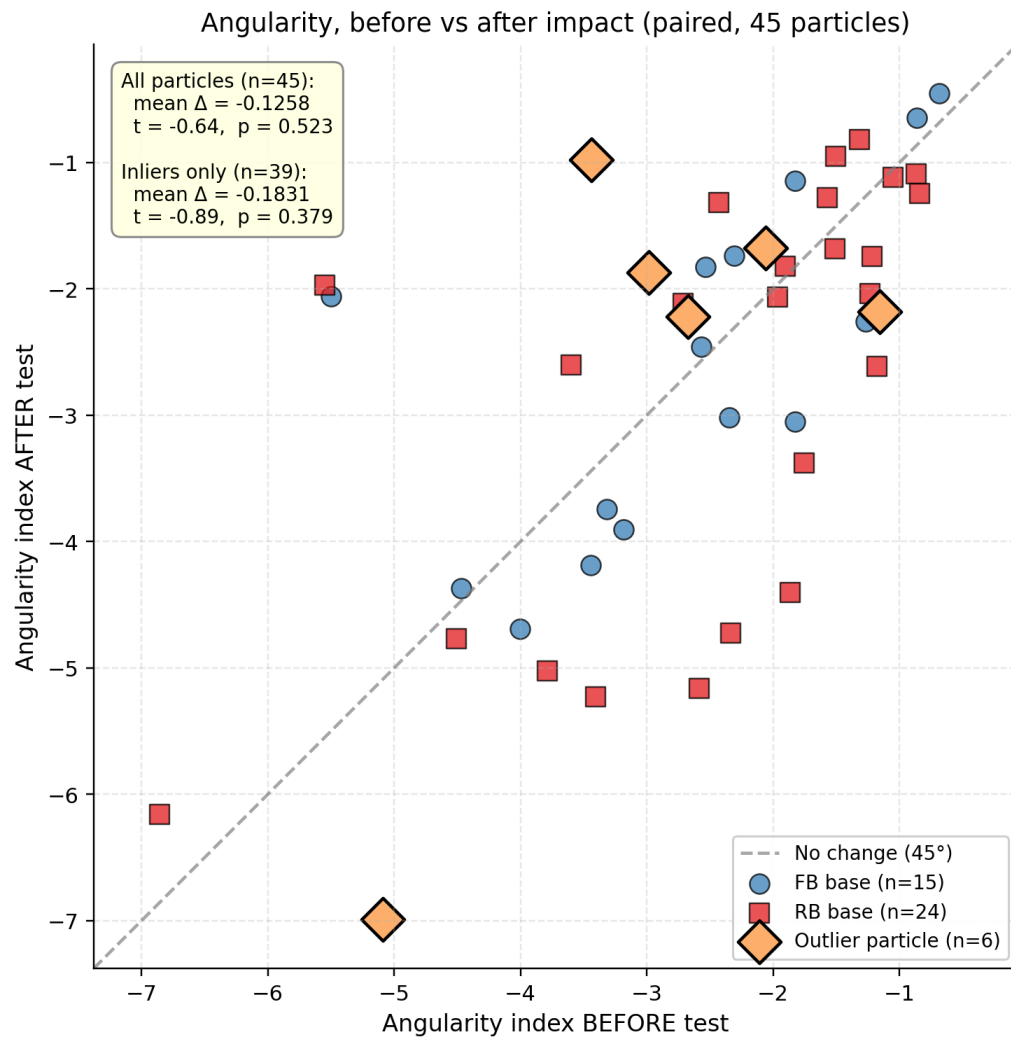


Figure 5.2. Angularity index before versus after the impact-loading test, for all 45 tracer particles. The angularity values are negative scalars; less-negative (closer to zero) values indicate smoother corners.

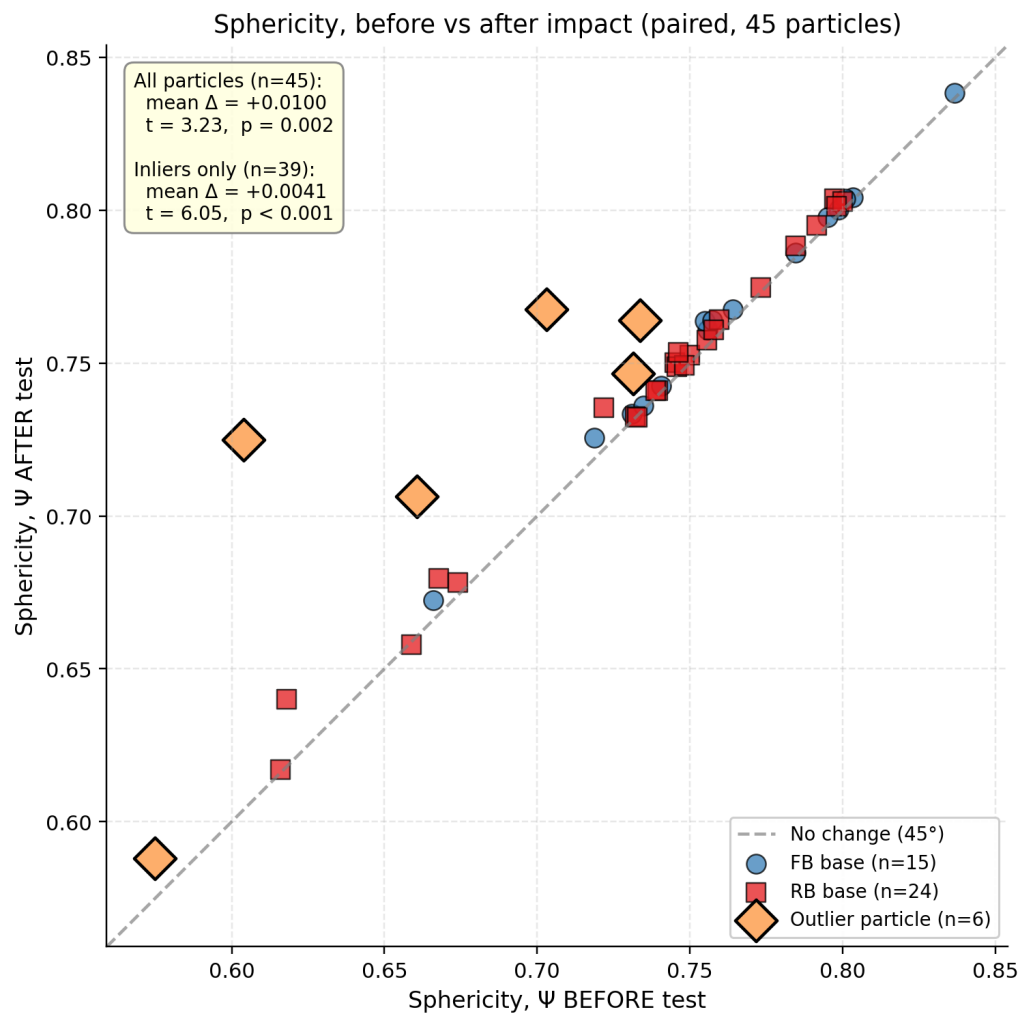


Figure 5.3. Sphericity before versus after the impact-loading test, for all 45 tracer particles.

Figure 5.4 reports the convexity $C = V/V_{ch}$ before versus after the test, with the same layout and statistics as for the sphericity. Convexity rises in parallel with sphericity, confirming that the particles become smoother and more compact as their corners are abraded.

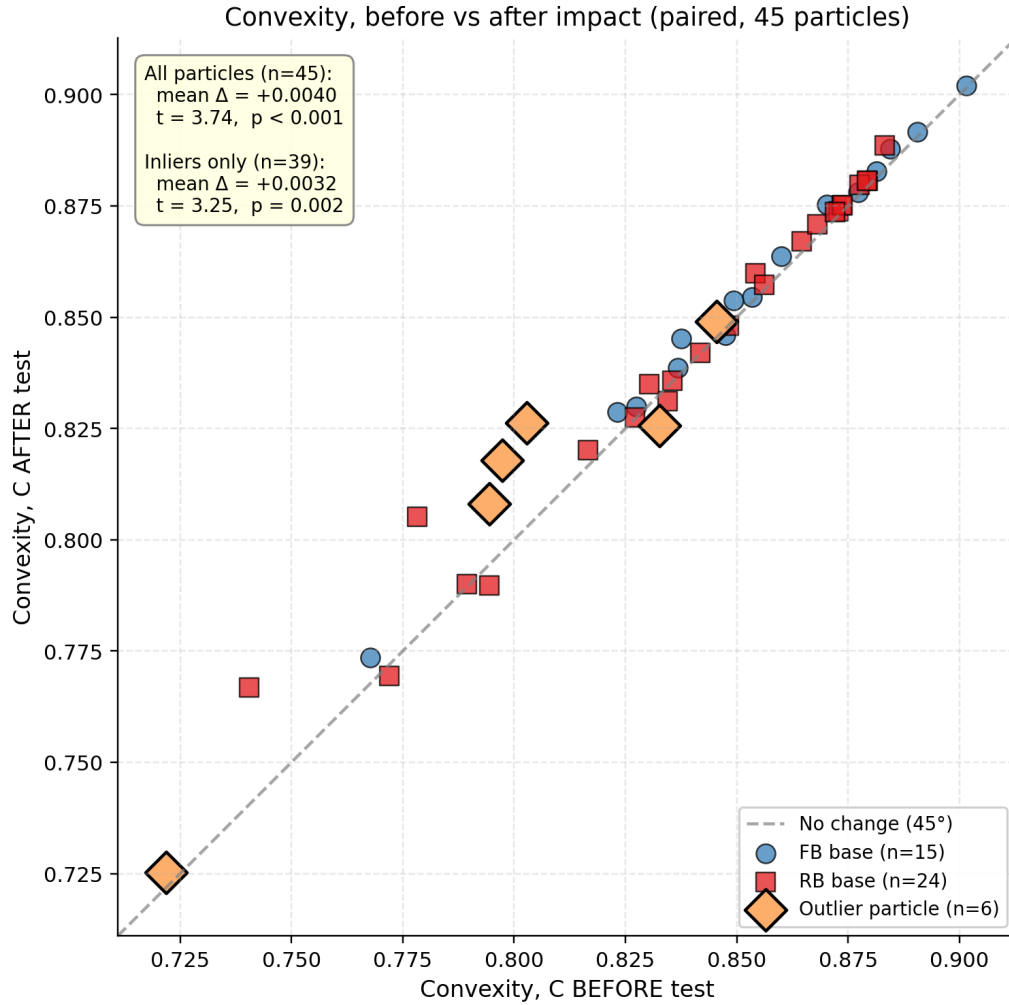


Figure 5.4. Convexity $C = V/V_{ch}$ before versus after the impact-loading test, for all 45 tracer particles. Layout and statistics as for Fig. 5.3.

The roundness response, by contrast, is mixed (Fig. 5.5). The mean paired change is +0.038 across all 45 particles ($t = 0.67$, $p = 0.51$), and +0.058 across the 39 inliers ($t = 1.00$, $p = 0.32$). Like angularity, roundness is sensitive to local corner curvature and reflects the selective nature of corner abrasion: most particles experience either a modest increase or a modest decrease in roundness, depending on which corner the impact-loading process abraded most. The sphericity-and-convexity rise reported above is the more reliable population-level diagnostic of the abrasion process; the angularity-and-roundness scatter reports the corresponding particle-level heterogeneity.

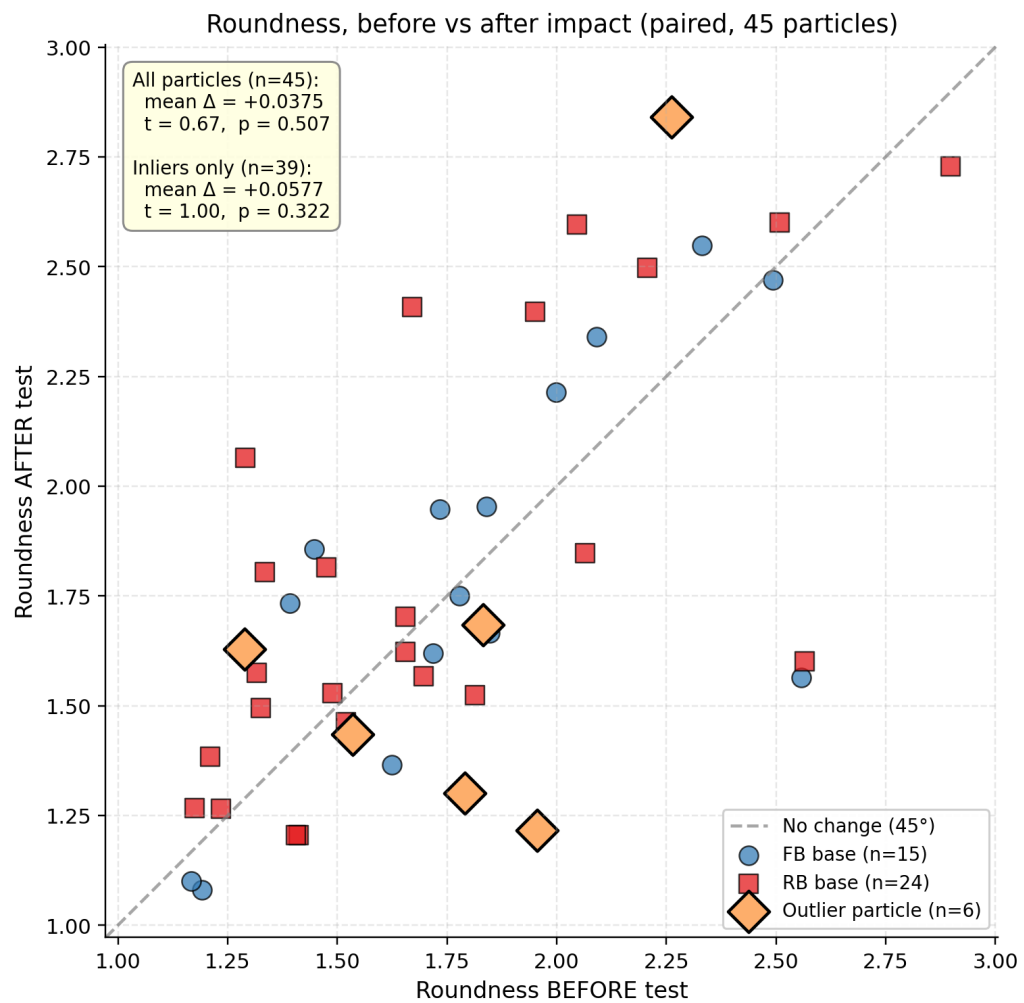


Figure 5.5. Roundness before versus after the impact-loading test, for all 45 tracer particles.

5.3 Discussion

The morphological script computes twenty-two descriptors per mesh, of which the four reported in Section 5.2 are the principal shape descriptors. To provide a global view of the pattern of mean change across all configurations and a wider set of descriptors, Fig. 5.6 collects the mean per-particle percentage change of fourteen descriptors across the nine tests in two side-by-side heat maps. Outlier particles are removed before the means are computed; the visualisation is therefore a Mode A (inlier) picture of the morphological response. The upper panel shows the size and primary shape descriptors, which have a comparatively narrow dynamic range (less than $\pm 2.5\%$); the lower panel shows the roundness and shape-ratio descriptors, which span a wider range (up to about $\pm 80\%$).

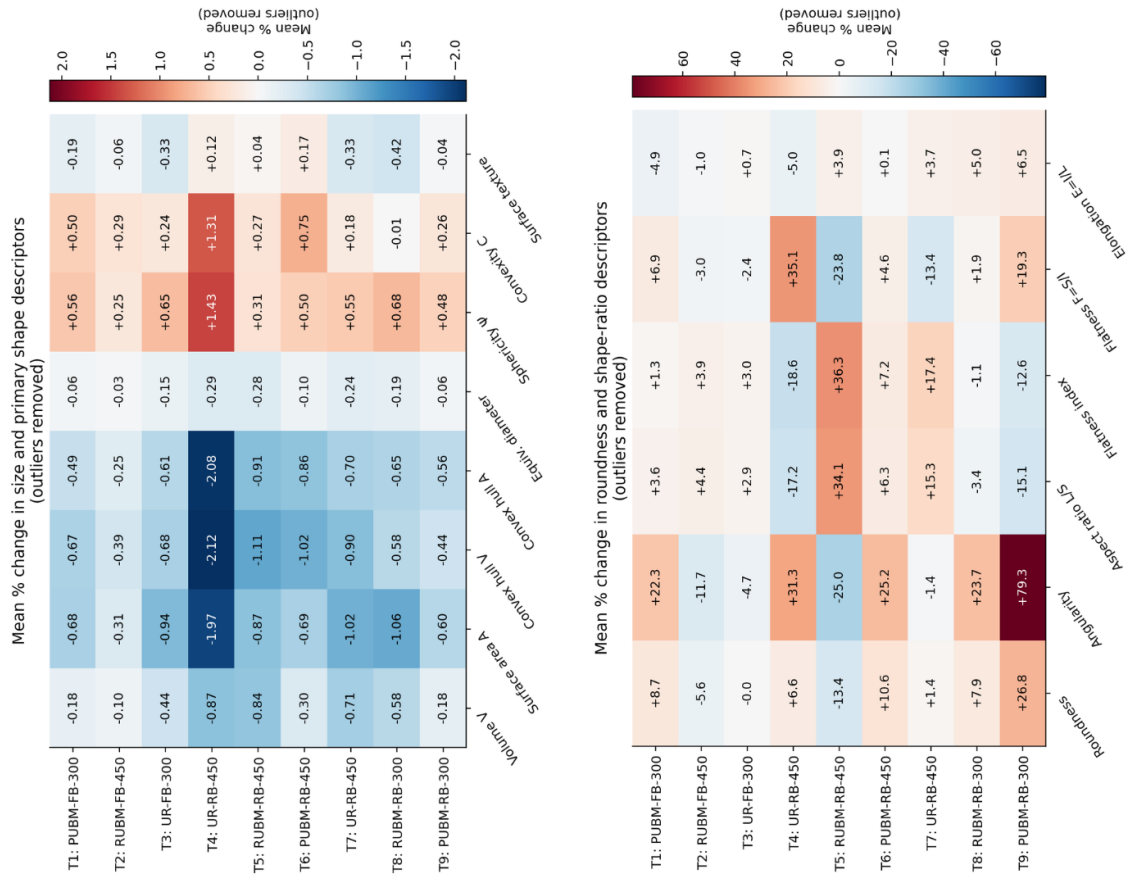


Figure 5.6. Heat maps of mean per-particle percentage change in fourteen morphological descriptors across the nine tests, with outlier particles removed. Upper panel: size and primary shape descriptors. Lower panel: roundness and shape-ratio descriptors.

Three patterns are visible in the upper panel of Fig. 5.6. First, all the size descriptors (volume, surface area, convex-hull volume, convex-hull surface, equivalent diameter) are uniformly negative across the nine configurations. The sign-uniformity is mechanistically reassuring: inlier particles do not gain mass under impact loading. The magnitude of the size loss tracks the ranking of Table 5.1, with the rigid-base 450 mm tests T4, T5, and T7

showing the largest size losses. Second, sphericity and convexity are uniformly positive across the nine configurations, consistent with the universal abrasion-driven rounding established in [Section 5.2](#). Third, the surface-texture-index column shows small but mixed-sign changes (between -0.42% and 0.17%), with no clear configuration-level pattern; this descriptor is sensitive to the highest-frequency mesh asperities and is dominated by scan-to-scan reconstruction noise rather than the impact-loading signal.

The lower panel of [Fig. 5.6](#) reports the roundness and shape-ratio descriptors, which exhibit a much larger dynamic range. The angularity column in particular spans from -25% at T5 to 79% at T9—a span of more than 100 percentage points across the nine configurations. The sign of the angularity change does not group obviously by base condition or by drop height: T1, T4, T6, T8, and T9 are all positive (sharper after the test, in the convention adopted here), while T2, T3, T5, and T7 are negative (smoother). The roundness column shows the same mixed-sign pattern, with the largest positive value at T9 (27%) and the largest negative value at T5 (-13%). The aspect-ratio and flatness columns are similarly heterogeneous. The mixed-sign pattern of these descriptors confirms the observation made in [Section 5.2](#) about angularity: corner abrasion is distributed selectively rather than uniformly across the corners of each particle, and the population-level mean of these locally sensitive descriptors is therefore not a reliable diagnostic of the configuration-level damage. The size descriptors and the global shape descriptors (sphericity and convexity) of the upper panel are the more robust diagnostics.

CHAPTER 6

Conclusions and Recommendations

This thesis set out to address a combined apparatus-and-characterisation gap in the current assessment of railway ballast degradation under drop-weight impact loading, specifically in the context of the RDSO 20–65 mm gradation envelope and the 22.5 t Indian Railways axle-load stress range. The programme comprised: the design, fabrication, and commissioning of a purpose-built large-scale drop-weight impact apparatus at Gati Shakti Vishwavidyalaya; the execution of a nine-configuration factorial test matrix crossing three pad conditions (UR / RUBM / PUBM) with two base conditions (FB / RB) and two impact energies (300 mm / 450 mm drop); the characterisation of bulk degradation through the Indian Railway Fouling Index FI_{IN} on the post-test gradations; the paired three-dimensional morphological characterisation of 45 individually tracked tracer particles through twenty-two descriptors before and after loading; and the statistical analysis of both characterisations individually and in synthesis. The present chapter states the principal conclusions of the programme, enumerates the specific novel contributions, identifies the limitations that bound the generalisability of the findings, and recommends concrete directions for follow-up research.

6.1 Conclusions

Six principal conclusions are drawn from the experimental programme and its analysis. They are stated here in the order in which they emerged during the research.

1. A large-scale drop-weight set-up was successfully developed at GSV, delivering consistent and reproducible impact energies representative of Indian Railways conditions.
2. Ballast shows rapid initial settlement that stabilises after about fifteen blows, confirming a twenty-blow test as appropriate. Higher energy and rigid bases cause greater settlement.
3. Fouling varies significantly with test conditions—highest under a rigid base, no pad, and high energy; lowest with a polyurethane pad, flexible base, and low energy.
4. Settlement and fouling measure different mechanisms (rearrangement versus breakage), so both are required for a full assessment.

5. Abrasion (Mode A): all particles undergo gradual corner abrasion, increasing sphericity and reducing size across all configurations. Fracture (Mode B): severe particle breakage occurs only under rigid-base conditions, owing to higher stress reflection.
6. FI_N captures bulk fouling effects, while particle morphology reveals damage mechanisms; both are essential for complete characterisation.

6.2 Novel contributions

6.2.1 Apparatus and method

- (a) A modified drop-weight impact apparatus for Indian ballast. A utility patent has been filed with the Indian Patent Office.
- (b) A standardised protocol calibrated to the Indian Railways stress window (275–523 kPa), recording per-blow peak force and five-milestone settlement—a full force-and-deformation history—versus the single post-test fines percentage of conventional Aggregate Impact Value testing.
- (c) A morphological-analysis method ([Appendix B](#)) that ingests paired pre/post STL meshes and exports the descriptors to a multi-sheet Excel workbook for direct comparison.

6.2.2 Scientific findings

- (a) Empirical demonstration that base condition (sand cushion versus rigid concrete) governs the selection between Mode A (diffuse abrasion) and Mode B (catastrophic fracture) under drop-weight impact loading of ballast, using a paired per-particle 3D morphological method.
- (b) A heat map of mean per-particle percentage change across fourteen descriptors for nine configurations: sign-uniformity in the size and primary-shape descriptors, heterogeneity in the shape-ratio descriptors, and sphericity and convexity as more robust diagnostics than angularity and roundness.
- (c) The first published nine-configuration factorial dataset pairing large-diameter drop-weight impact loading (RDSO gradation, Indian Railways stress levels) with per-particle 3D morphology—a reference for validating discrete-element models and machine-learning degradation predictors.

6.3 Limitations of the study

The following limitations bound the generalisability of the conclusions and should be recorded in any reading of the findings.

1. **Sample size per configuration.** Five tracer particles supports the pooled Mode A significance test and Mode B detection, but not per-configuration ANOVA or tight Mode B probability estimates. Expanding to 10–20 tracers is the single most effective follow-up improvement.
2. **Single parent rock type.** Only basalt from one Vadodara quarry was tested. Parent rock affects both the Mode A abrasion rate and the Mode B fracture threshold, so generalisation requires extension to another RDSO-approved rock.
3. **Surface morphology only.** The EinScan structured-light scanner cannot detect sub-surface micro-cracks, incipient fracture planes, or mineralogical alteration. Mode A particles may carry latent damage that accelerates Mode B fracture; X-ray CT would be required to detect this.

6.4 Recommendations for future work

The following concrete directions for follow-up research are identified, in rough order of increasing scope. Each would usefully build on the methodological platform established by the present thesis.

1. **Expanded tracer-count study.** Repeat the present nine-configuration programme with 15–20 tracer particles per configuration to support per-configuration ANOVA and tighter estimates of Mode B event probabilities.
2. **Multi-rock-type extension.** Extend the programme to granite and quartzite ballast, using the same nine-configuration matrix and the same per-particle morphology pipeline, to map the parent-rock dependence of the Mode A / Mode B crossover.
3. **AI/ML-based prediction of degradation behaviour.** Use the per-particle paired morphological dataset to train supervised machine-learning models. A trained surrogate would allow rapid screening of pad–base combinations without running the full impact campaign, and feature-importance analysis would identify which morphological descriptors most strongly govern degradation outcomes.
4. **Cyclic-plus-impact combined loading.** Develop a composite loading method that applies cyclic-triaxial loading (mimicking ordinary traffic) interrupted periodically by drop-weight impact events (mimicking transition-zone hot-spots), and report the corresponding Mode A / Mode B evolution. This is the closest laboratory analogue to real service loading.

References

- Alabbasi, Y. and Hussein, M. (2021). Geomechanical modelling of railroad ballast: A review. *Archives of Computational Methods in Engineering*, 28(3):815–839.
- American Railway Engineering and Maintenance-of-Way Association (2020). *Manual for Railway Engineering – Chapter 1: Roadway and Ballast*. AREMA.
- Anbazzhagan, P., Lijun, S., Bharatha, T. P., and Pratibha, R. (2012). Model track studies on fouled ballast using seismic surface wave method. *Journal of Applied Geophysics*, 86:108–115.
- Angelidakis, V., Nadimi, S., and Utili, S. (2021). SHape Analyser for Particle Engineering (SHAPE): Seamless characterisation and simplification of particle morphology from imaging data. *Computer Physics Communications*, 265:107983.
- Bian, X., Cai, W., Luo, Z., Zhao, C., and Chen, Y. (2023). Image-aided analysis of ballast particle movement along a high-speed railway. *Engineering*, 27:161–177.
- Bishop, A. W. (1969). The influence of system compressibility on the observed pore-pressure response to an undrained change in stress in saturated rock. *Géotechnique*, 19(4):514–531.
- British Standards Institution (2010). *Tests for Mechanical and Physical Properties of Aggregates – Part 2: Methods for the Determination of Resistance to Fragmentation (BS EN 1097-2)*. BSI.
- Bureau of Indian Standards (1963). *Methods of Test for Aggregates for Concrete, Part IV: Mechanical Properties (IS:2386 Part IV)*. BIS.
- Chen, C., Zhang, H., Yang, J., and Wu, Z. (2026). Particle breakage and gradation evolution in ballast–rubber mixture under impact loading. *Géotechnique Letters*. Advance online publication.
- Cundall, P. A. and Strack, O. D. L. (1979). A discrete numerical model for granular assemblies. *Géotechnique*, 29(1):47–65.
- Dawson-Haggerty, M. (2019). trimesh: Python library for loading and using triangular meshes. PyPI.
- Esveld, C. (2001). *Modern Railway Track*. MRT-Productions, 2nd edition.

- Ferreira, L. and Indraratna, B. (2023). Image-based morphological characterisation of railroad ballast degradation under impact loading. *Transportation Geotechnics*, 39:100938.
- Gundavaram, D. and Hussaini, S. K. K. (2023). Application of elastomeric polyurethane in performance improvement of rail ballast subjected to cyclic loading. *Journal of Materials in Civil Engineering*, 35(4):04023019.
- Guo, Y., Markine, V., Song, J., and Jing, G. (2018). Ballast degradation: Effect of particle size and shape using Los Angeles abrasion test and image analysis. *Construction and Building Materials*, 169:414–424.
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585(7825):357–362.
- Huang, H., Tutumluer, E., and Dombrow, W. (2009). Laboratory characterization of fouled railroad ballast behavior. *Transportation Research Record*, 2117(1):93–101.
- Indraratna, B., Salim, W., and Rujikiatkamjorn, C. (2011). *Advanced Rail Geotechnology – Ballasted Track*. CRC Press.
- Jayasuriya, C., Indraratna, B., and Ngo, T. N. (2019). Experimental study to examine the role of under sleeper pads for improved performance of ballast under cyclic loading. *Transportation Geotechnics*, 19:61–73.
- Koohmishi, M. and Azarhoosh, A. (2021). Degradation of crumb rubber modified railway ballast under impact loading considering aggregate gradation and rubber size. *Canadian Geotechnical Journal*, 58(2):398–410.
- Koohmishi, M. and Azarhoosh, A. (2022). Assessment of performance of recycled ballast aggregate subjected to drop-weight impact loading considering effect of crumb rubber incorporation. *Journal of Materials in Civil Engineering*, 34(5):04022057.
- Koohmishi, M. and Guo, Y. (2023). Machine learning approach to railway ballast degradation prognosis considering crumb rubber modification and parent rock strength. *Construction and Building Materials*, 409:133985.
- Koohmishi, M. and Palassi, M. (2020). Degradation of railway ballast under impact loading considering the morphological properties of aggregate. *Transportation Geotechnics*, 25:100398.
- Krumbein, W. C. (1941). Measurement and geological significance of shape and roundness of sedimentary particles. *Journal of Sedimentary Research*, 11(2):64–72.
- Kwunjai, S., Jitsangiam, P., and Somsri, T. (2024). Morphological analysis of ballast particles: Characterization and simplified analysis using imaging data. *IOP Conference Series: Earth and Environmental Science*, 1332:012016.

- Kwunjai, S., Somsri, T., Jitsangiam, P., Binaree, T., Qian, Y., and Jing, G. (2023). Characterization of deteriorated railway ballast morphological changes using 3D scanning and supervised machine learning data analytics. *Construction and Building Materials*, 398:132445.
- Li, D. and Davis, D. (2005). Transition of railroad bridge approaches. *Journal of Geotechnical and Geoenvironmental Engineering*, 131(11):1392–1398.
- Lichtberger, B. (2005). *Track Compendium: Formation, Permanent Way, Maintenance, Economics*. Eurailpress.
- Lim, W. L. and McDowell, G. R. (2005). Discrete element modelling of railway ballast. *Granular Matter*, 7(1):19–29.
- Lim, W. L., McDowell, G. R., and Collop, A. C. (2004). The application of Weibull statistics to the strength of railway ballast. *Granular Matter*, 6(4):229–237.
- Marsal, R. J. (1967). Large-scale testing of rockfill materials. *Journal of the Soil Mechanics and Foundations Division*, 93(2):27–43.
- Navaratnarajah, S. K. and Indraratna, B. (2017). Use of rubber mats to improve the deformation and degradation behavior of rail ballast under cyclic loading. *Journal of Geotechnical and Geoenvironmental Engineering*, 143(6):04017015.
- Ngo, N. T., Indraratna, B., and Rujikiatkamjorn, C. (2019). Improved performance of ballasted tracks under impact loading by recycled rubber mats. *Transportation Geotechnics*, 20:100239.
- Nimbalkar, S., Indraratna, B., Dash, S. K., and Christie, D. (2012). Improved performance of railway ballast under impact loads using shock mats. *Journal of Geotechnical and Geoenvironmental Engineering*, 138(3):281–294.
- Qian, Y., Boler, H., Moaveni, M., Tutumluer, E., Hashash, Y. M. A., and Ghaboussi, J. (2014). Characterizing ballast degradation through Los Angeles abrasion test and image analysis. *Transportation Research Record*, 2448(1):142–151.
- Remennikov, A. M. and Kaewunruen, S. (2008). A review of loading conditions for railway track structures due to train and track vertical interaction. *Structural Control and Health Monitoring*, 15(2):207–234.
- Research Designs and Standards Organisation (2023). *Specification for Track Ballast (IS/RDSO-GE/0001:2023)*. Ministry of Railways, Government of India.
- Selig, E. T. and Waters, J. M. (1994). *Track Geotechnology and Substructure Management*. Thomas Telford.
- Sol-Sánchez, M., Thom, N. H., Moreno-Navarro, F., Rubio-Gámez, M. C., and Airey, G. D. (2015). A study into the use of crumb rubber in railway ballast. *Construction and*

- Building Materials*, 75:19–24.
- Sun, Y., Indraratna, B., and Nimbalkar, S. (2014). Three-dimensional characterisation of particle size and shape for ballast. *Géotechnique Letters*, 4(3):197–202.
- Tordesillas, A. (2007). Force chain buckling, unjamming transitions and shear banding in dense granular assemblies. *Philosophical Magazine*, 87(32):4987–5016.
- Tutumluer, E. (2013). Practices for unbound aggregate pavement layers (nchrp synthesis 445). Technical report, Transportation Research Board.
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272.
- Wadell, H. (1932). Volume, shape, and roundness of rock particles. *Journal of Geology*, 40(5):443–451.
- Werkmeister, S., Dawson, A. R., and Wellner, F. (2001). Permanent deformation behavior of granular materials and the shakedown concept. *Transportation Research Record*, 1757(1):75–81.
- Zhang, F., Chang, J., and Feng, H. (2023). Laboratory study on degradation of ballast mixed with crumb rubber under impact loads. *International Journal of Rail Transportation*, 11(5):767–789.

CHAPTER A

Tabulated Data Excerpts

This appendix reproduces tabulated excerpts from the principal laboratory data files referenced throughout the thesis. The full raw data—including all 45 paired before–after scans and the twenty-two descriptors per mesh—is archived in the two Excel workbooks produced by the pipeline of [Appendix B](#) and is available from the authors on reasonable request.

A.1 Test matrix — full specification

[Table A.1](#) gives the complete nine-configuration test matrix with the measured peak force, kinetic energy, and nominal contact stress, together with the legacy low-height (LH, 300 mm) and high-height (HH, 450 mm) drop labels.

Table A.1. Complete nine-configuration test matrix with measured peak force, kinetic energy, nominal contact stress, and the legacy systematic test labels.

Test	Systematic label	Pad	Base	h (mm)	K_e (J)	Peak force (kN)	σ (kPa)
T1	PUBM-FB-LH	PUBM	FB	300	147	19.4	275
T2	RUBM-FB-LH	RUBM	FB	300	147	19.4	275
T3	UR-FB-LH	UR	FB	300	147	19.4	275
T4	UR-RB-LH	UR	RB	300	147	19.4	275
T5	RUBM-RB-HH	RUBM	RB	450	221	37.0	523
T6	PUBM-RB-HH	PUBM	RB	450	221	37.0	523
T7	UR-RB-HH	UR	RB	450	221	37.0	523
T8	RUBM-RB-LH	RUBM	RB	300	147	19.4	275
T9	PUBM-RB-LH	PUBM	RB	300	147	19.4	275

A.2 Per-particle paired morphology — summary table

[Table A.2](#) gives the per-configuration Mode A summary of the principal morphological change descriptors. Mode B outliers are excluded; ΔV , $\Delta\Psi$, and ΔC are reported as percentage relative change, and n is the Mode A sample size in each configuration.

Table A.2. Per-configuration Mode A summary of the principal morphological change descriptors. Mode B outliers excluded.

Test	Label	n (Mode A)	Mode B events	Mean $\% \Delta V$	Mean $\% \Delta \Psi$	Mean $\% \Delta C$
T1	PUBM-FB-LH	5	0	−0.18	0.56	0.50
T2	RUBM-FB-LH	5	0	−0.10	0.25	0.29
T3	UR-FB-LH	5	0	−0.44	0.65	0.24
T4	UR-RB-LH	3	2	−0.87	1.43	1.31
T5	RUBM-RB-HH	4	1	−0.84	0.31	0.27
T6	PUBM-RB-HH	4	1	−0.30	0.50	0.75
T7	UR-RB-HH	4	1	−0.72	0.55	0.18
T8	RUBM-RB-LH	4	1	−0.58	0.68	−0.01
T9	PUBM-RB-LH	5	0	−0.18	0.48	0.26
All (pooled)		39	6	−0.47	0.56	0.40

A.3 Complete post-test sieve data

Table A.3 gives the percentage retained on each sieve, post-test, for the nine configurations. Rows are sieve sizes; columns are tests (the full configuration of each test is given in Table A.1). Values are in weight percent of the post-test specimen mass.

Table A.3. Percentage retained on each sieve, post-test, for the nine configurations. The lower block gives the specimen total mass and the derived fouling fractions.

Sieve (mm)	T1	T2	T3	T4	T5	T6	T7	T8	T9
63	7.18	8.34	8.44	12.11	8.01	6.45	8.35	11.00	8.63
40	9.15	8.09	6.44	12.25	12.15	10.85	9.62	13.02	9.26
31.5	2.11	2.65	2.15	2.44	1.55	5.75	1.50	3.44	2.53
20	0.210	0.375	0.500	0.675	0.270	0.200	0.125	0.250	0.270
16	0.010	0.110	0.050	0.090	0.036	0.100	0.080	0.010	0.033
12.5	0.004	0.043	0.037	0.120	0.021	0.015	0.150	0.022	0.052
10	0.007	0.026	0.030	0.070	0.030	0.011	0.065	0.010	0.028
4.75	0.012	0.054	0.044	0.130	0.071	0.021	0.130	0.041	0.040
2.36	0.015	0.009	0.019	0.038	0.032	0.009	0.047	0.019	0.016
1.18	0.010	0.009	0.014	0.026	0.025	0.006	0.046	0.014	0.012
0.6	0.025	0.005	0.025	0.014	0.015	0.004	0.037	0.010	0.006
0.075	0.022	0.013	0.022	0.035	0.050	0.011	0.083	0.040	0.024
Pan	0.005	0.004	0.005	0.013	0.011	0.004	0.024	0.010	0.004
Total (g)	18.76	19.73	17.78	28.01	22.27	23.43	20.26	27.89	20.91
$P_{0.075}$ (%)	0.12	0.07	0.12	0.12	0.22	0.05	0.41	0.14	0.11
P_{20} (%)	0.50	1.09	0.95	1.16	1.01	0.26	2.28	0.62	0.77
FI_{IN} (%)	0.62	1.16	1.07	1.28	1.23	0.31	2.69	0.76	0.88

CHAPTER B

Python Morphological-Analysis Script

This appendix records the Python morphological-analysis script. The script reads STL meshes from a pre-test folder and a post-test folder, performs the four-check mesh quality-control gate, computes the twenty-two morphological descriptors, pairs the pre-test and post-test meshes by filename, computes the per-particle changes, and exports the complete dataset to a multi-sheet Excel workbook.

B.1 Software environment

- Python 3.10 or later (tested on 3.12).
- `trimesh` $\geq$ 4.0 — STL input, mesh-integrity checks, convex hull.
- `numpy` $\geq$ 1.24 — numerical operations on descriptor arrays.
- `pandas` $\geq$ 2.0 — DataFrame handling for the paired dataset.
- `numpy-stl` — legacy STL-format compatibility layer.

Installation is via the `pip` toolchain:

```
pip install trimesh numpy pandas openpyxl scipy numpy-stl
```

B.2 Pipeline script — `ballast_analyzer.py`

The complete pipeline script is archived as `ballast_analyzer.py`. The script is approximately 450 lines of commented Python; the core descriptor-extraction loop is reproduced below in abridged form.

```
1  import os, trimesh, numpy as np, pandas as pd
2
3  # ----- Four-check quality gate -----
4  def mesh_ok(mesh):
5      return (mesh.is_watertight
6              and mesh.body_count == 1
7              and mesh.volume > 10_000
8              and mesh.volume < 250_000)
9
10 # ----- Descriptor extraction -----
```



```

11 def extract_descriptors(stl_path):
12     m = trimesh.load(stl_path)
13     if not mesh_ok(m):
14         raise ValueError(f"{stl_path}: mesh QC failed")
15     V, A = m.volume, m.area
16     ch = m.convex_hull
17     V_ch, A_ch = ch.volume, ch.area
18     # Principal dimensions from oriented bounding box
19     obb = m.bounding_box_oriented.extents # [S, I, L]
20     S, I, L = sorted(obb)
21     # Sphericity (Wadell)
22     Psi = (np.pi ** (1/3)) * ((6*V) ** (2/3)) / A
23     # Convexity
24     C = V / V_ch
25     # Form descriptors
26     F, E, AR = S/I, I/L, L/S
27     FI = (L + I) / (2*S)
28     # Surface texture
29     T = A / A_ch
30     D_eq = (6*V/np.pi) ** (1/3)
31     return dict(V=V, A=A, D_eq=D_eq, S=S, I=I, L=L,
32                V_ch=V_ch, A_ch=A_ch, Psi=Psi, C=C,
33                F=F, E=E, AR=AR, FI=FI, T=T)
34
35 # ----- Pair before & after, write workbook -----
36 def run(before_dir, after_dir, out_xlsx):
37     rows_b, rows_a = [], []
38     for fn in sorted(os.listdir(before_dir)):
39         if not fn.endswith(".stl"):
40             continue
41         d_b = extract_descriptors(os.path.join(before_dir, fn))
42         d_a = extract_descriptors(os.path.join(after_dir, fn))
43         d_b["Particle"] = fn.replace(".stl", "")
44         d_a["Particle"] = fn.replace(".stl", "")
45         rows_b.append(d_b)
46         rows_a.append(d_a)
47     df_b = pd.DataFrame(rows_b).set_index("Particle")
48     df_a = pd.DataFrame(rows_a).set_index("Particle")
49     df_delta = df_a - df_b
50     df_pct = 100 * df_delta / df_b
51     with pd.ExcelWriter(out_xlsx) as xw:
52         df_b.to_excel(xw, "Before Test")
53         df_a.to_excel(xw, "After Test")
54         df_delta.to_excel(xw, "Changes (After-Before)")
55         df_pct.to_excel(xw, "Percent Changes")

```

Listing B.1. Core of the morphological-analysis pipeline (abridged).